



Signals and Systems EE252, Spring 2026

Instructor: Tariq Mumtaz

Semester week # 14





Course Learning Outcome (CLOs):

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)



Roadmap till Semester end:

Frequency Domain Analysis of CT/DT Signals & Systems:

Z- Transform (Wk 7 – 11)

Laplace Transform (Wk 12 – 13)

→ Fourier Transform (Wk 14 – 15)

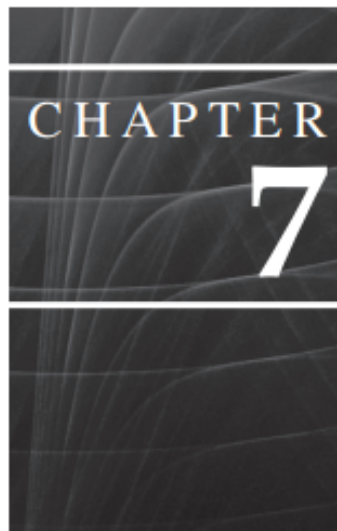
Project presentations (Wk16)



Learning outcome of Wk#13 session:

Frequency Domain Analysis using Fourier Transform(Chapter 07)

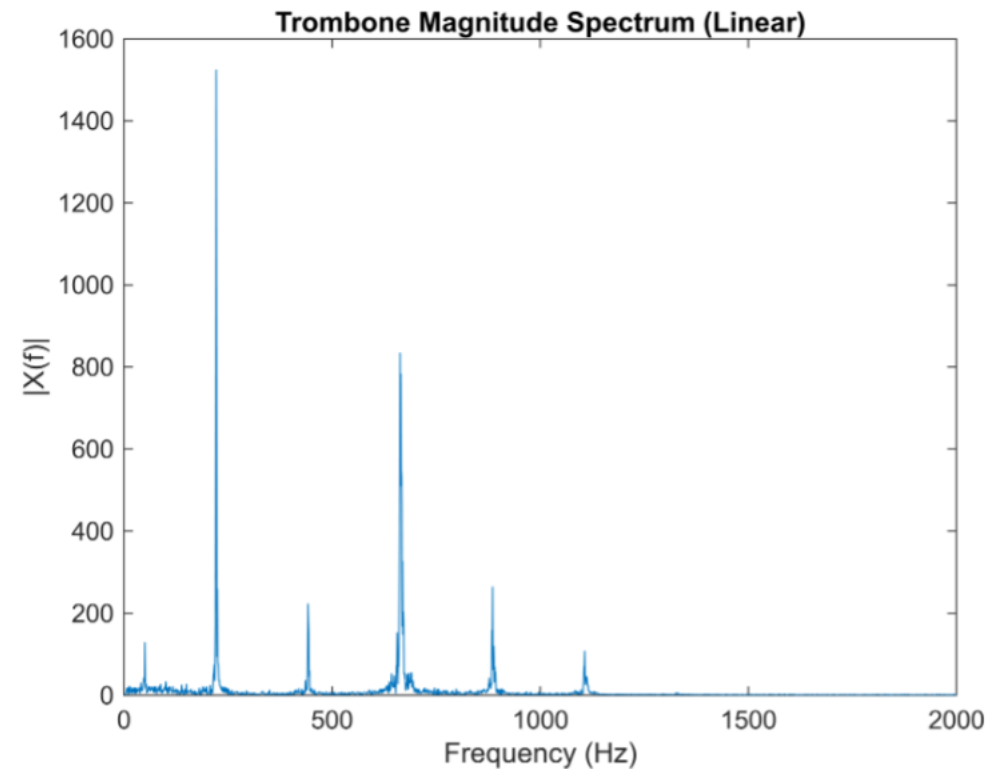
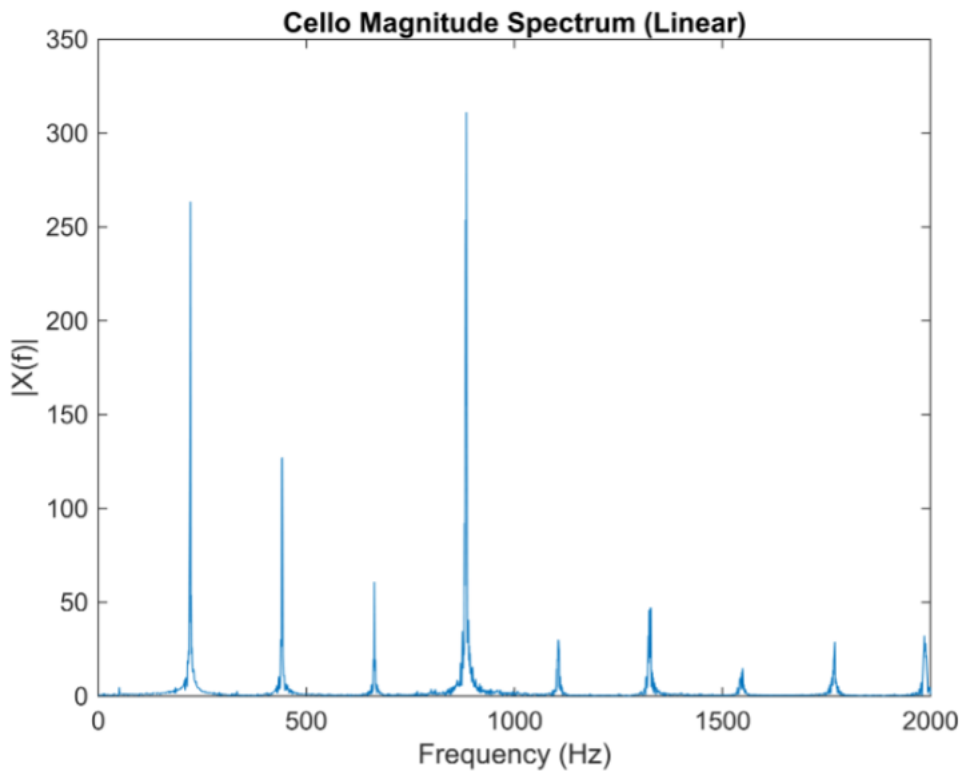
- CT Signal Representation by FT
- FT of basic signals
- FT properties
- FT of LTIC system



CONTINUOUS-TIME SIGNAL ANALYSIS: THE FOURIER TRANSFORM

Fourier Transform:

SnS Project – Frequency Analysis of Cello and Trombone using FFT

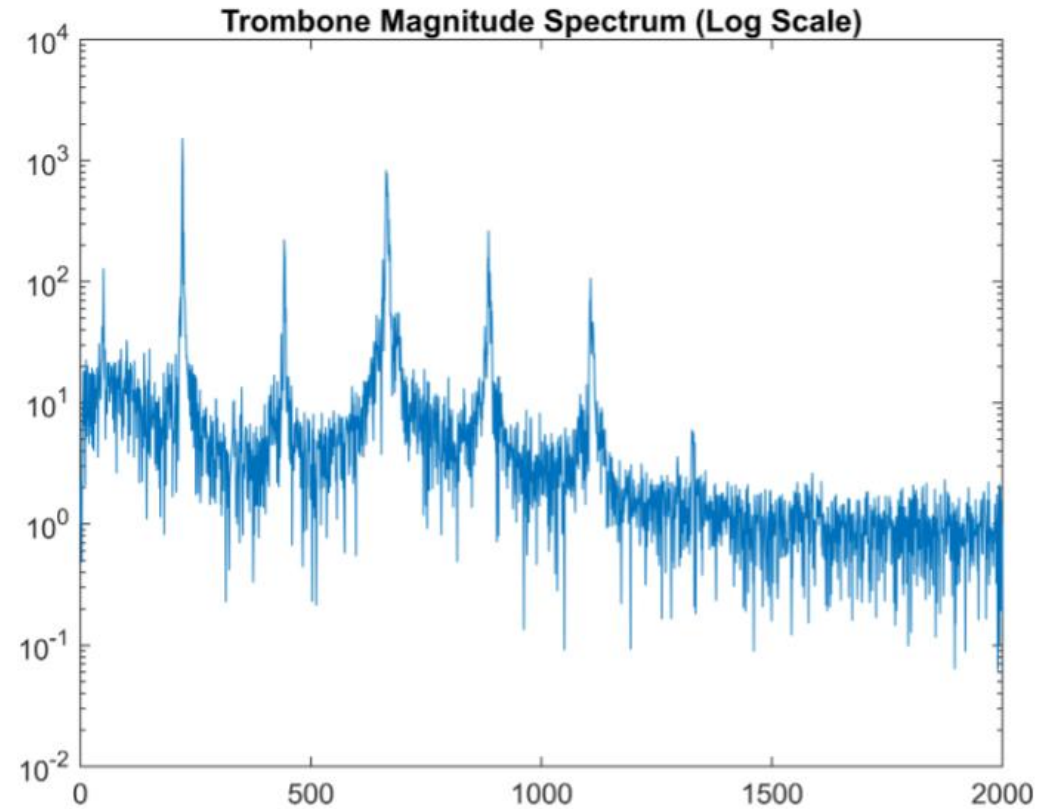
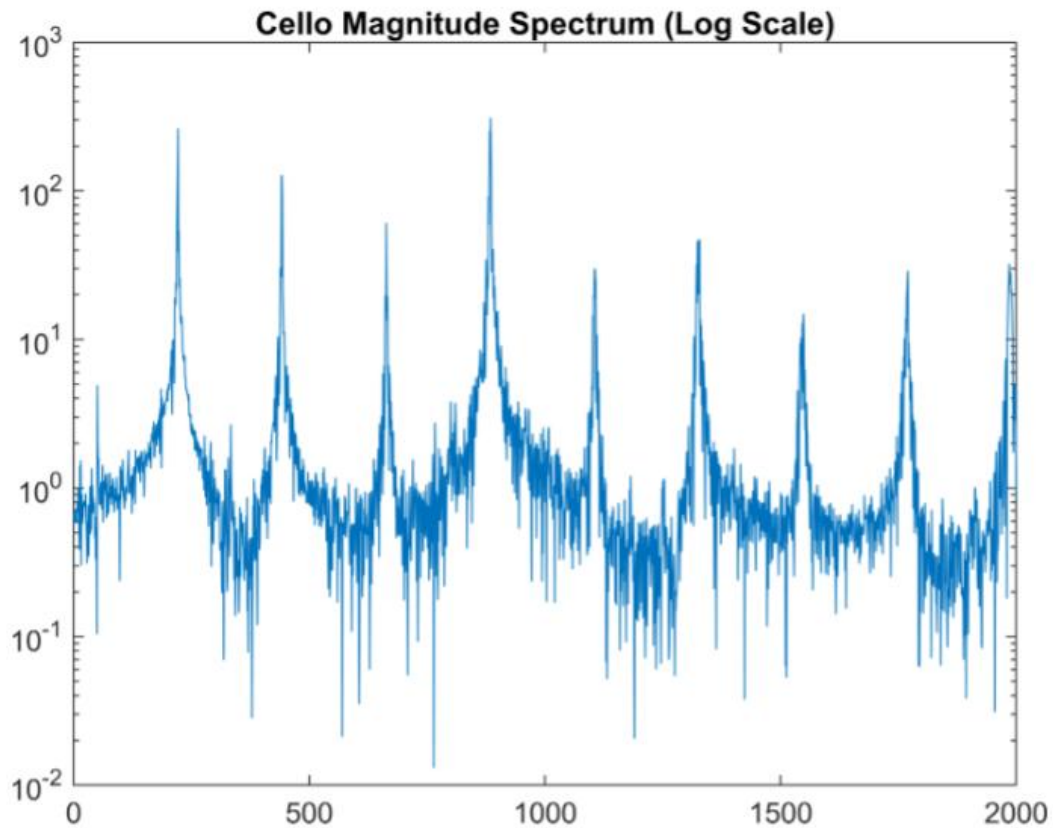


Insights from FT Plot?

Spikes @ Different frequencies – does it look like the impulse signal?

Fourier Transform:

SnS Project – Frequency Analysis of Cello and Trombone using FFT

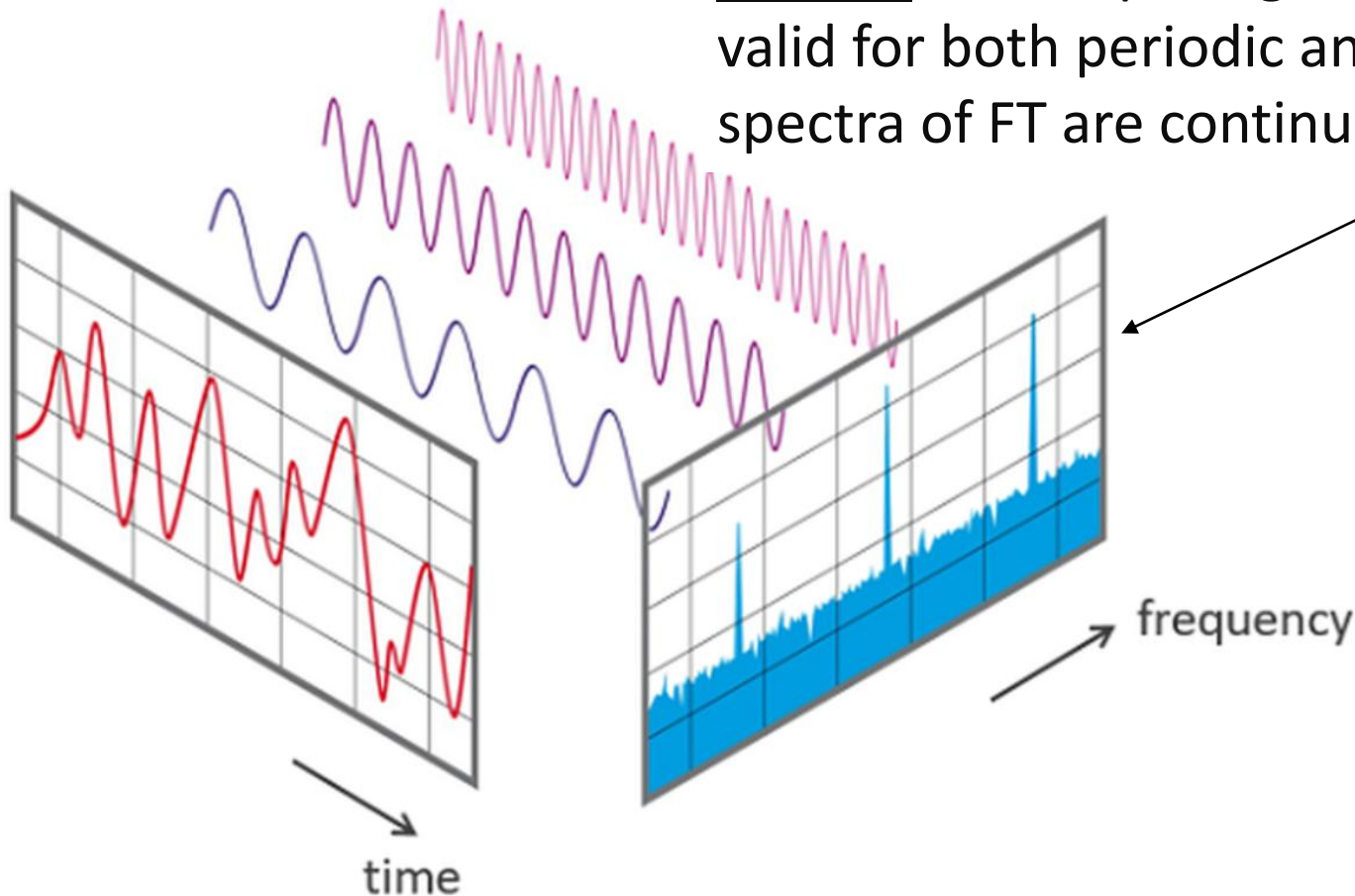


Still impulse? If not, then what is this signal that's used to decompose complex music instrument audio signals or any arbitrary signal? All this we explore in FT analysis

Fourier Transform:

Fourier Transform: It represents a signal in the frequency domain, decomposing it into its component frequencies. FT is valid for both periodic and aperiodic time domain signals. The spectra of FT are continuous in the frequency domain

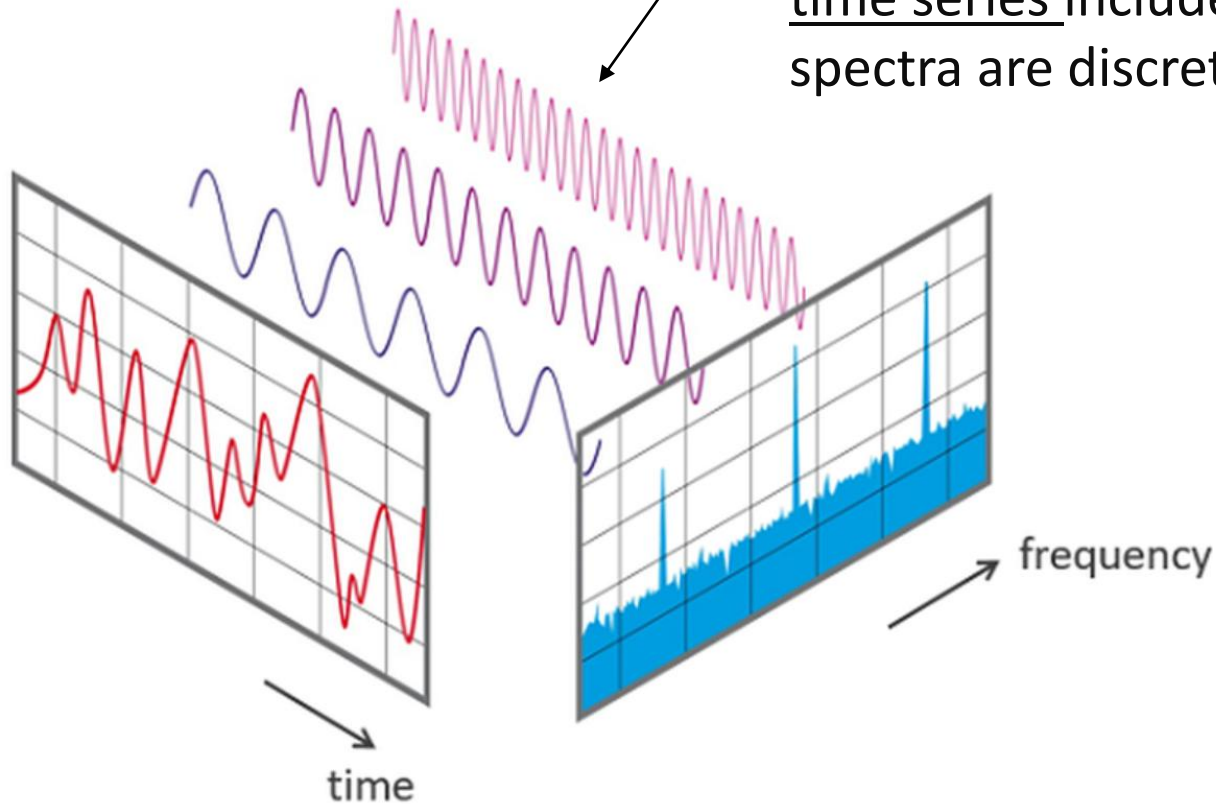
WK# 14



*FT has two types: CTFT and DTFT
CTFT: SnS course
DTFT: DSP course (FFT is an
algorithm to compute DTFT
efficiently.)*

Fourier Transform:

Fourier Series: It represents ONLY a periodic time –domain signal as the sum of sinusoidal functions (sine and cosine) or complex exponentials. The decomposition of a periodic signal in terms of a time series includes a fundamental frequency and its harmonics. Its spectra are discrete in the frequency domain.



WK# 15

The last topic of the SnS course

Fourier Transform:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

- Fourier Transform helps to write $x(t)$ as a combination of complex sinusoids $e^{j\omega t}$. Here, $e^{j\omega t}$ combines cosine and sine (Euler's Formula).
- $X(\omega)$ tells what frequencies are present in $x(t)$ or FT decompose $x(t)$ into a range of frequencies

$$X(\omega) = \text{Re}\{X(\omega)\} + j \text{Im}\{X(\omega)\}$$

$$X(\omega) = |X(\omega)| e^{j\angle X(\omega)}$$

- $|X(\omega)|$ tells how strong (amplitude) the frequency is
- $\angle X(\omega)$ tells how the frequency is shifted in phase

$X(\omega)$ stores the magnitudes and phases of all sinusoids needed to reconstruct $x(t)$

Fourier Transform:

FT

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

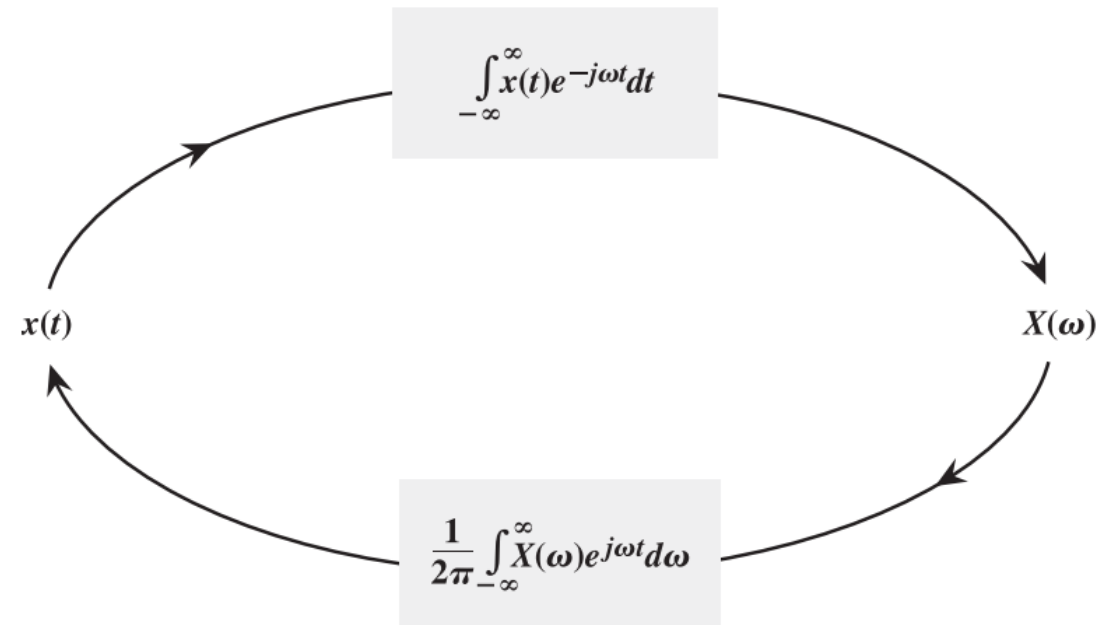
Invers FT

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

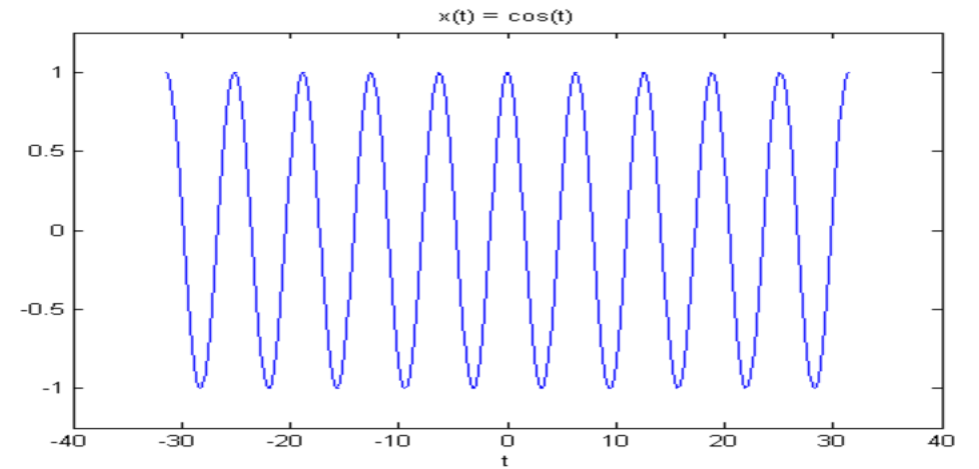
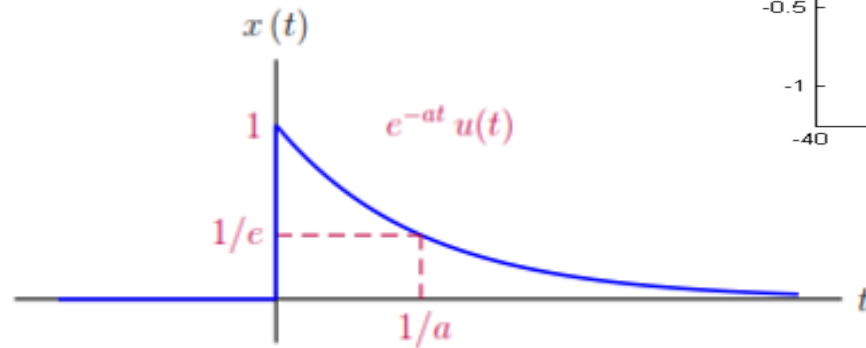
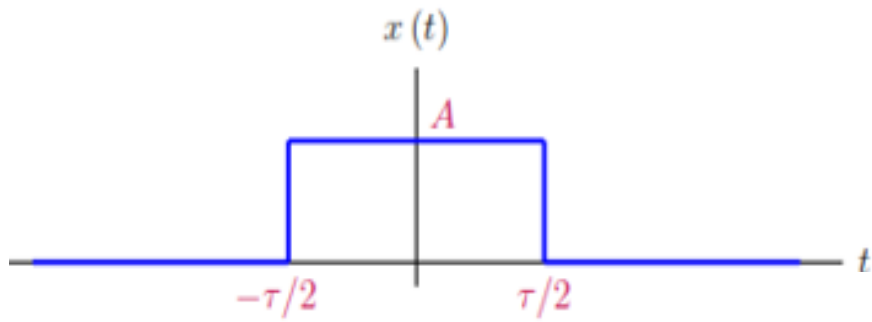
FT pair notation

$$X(\omega) = \mathcal{F}[x(t)] \quad \text{and} \quad x(t) = \mathcal{F}^{-1}[X(\omega)]$$

$$x(t) \Longleftrightarrow X(\omega)$$



Fourier Transform of Basic Signals



Fourier Transform: The $\delta(t)$ signal

EXAMPLE 7.3 Fourier Transform of the Dirac Delta Function

Find the Fourier transform of the unit impulse $\delta(t)$.

Using the sampling property of the impulse [Eq. (1.11)], we obtain

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1 \quad \text{and} \quad \delta(t) \longleftrightarrow 1$$

Figure 7.11 shows $\delta(t)$ and its spectrum.

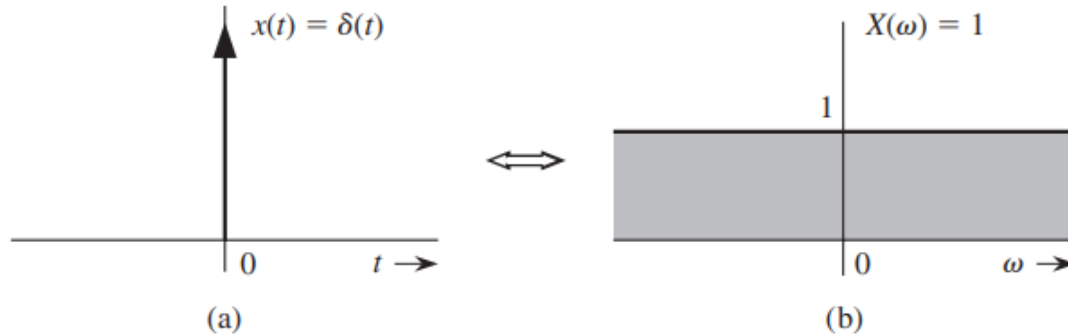


Figure 7.11 (a) Unit impulse and (b) its Fourier spectrum.

A very narrow signal in the time domain is Infinite in the frequency domain.

Fourier Transform: The $\delta(t)$ signal

EXAMPLE 7.4 Inverse Fourier Transform of the Dirac Delta Function

Find the inverse Fourier transform of $\delta(\omega)$.

On the basis of Eq. (7.10) and the sampling property of the impulse function,

$$\mathcal{F}^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi}$$

Therefore,

$$\frac{1}{2\pi} \Longleftrightarrow \delta(\omega) \quad \text{and} \quad 1 \Longleftrightarrow 2\pi \delta(\omega)$$

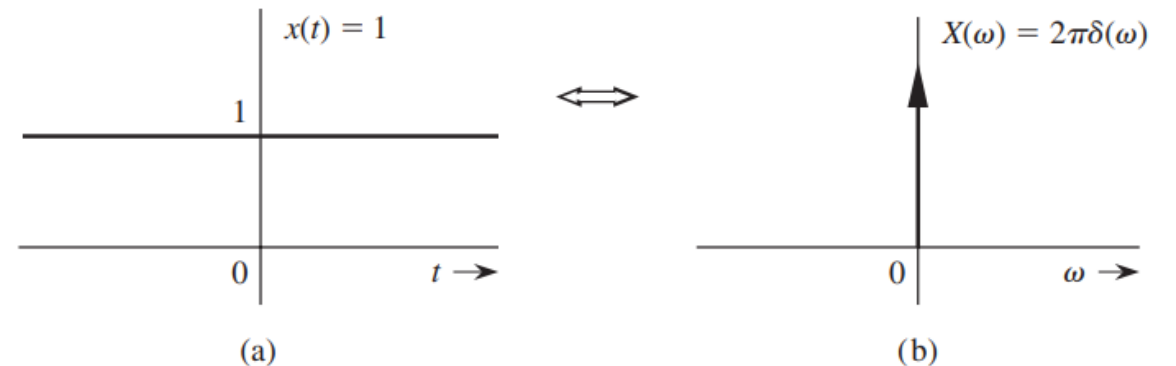


Figure 7.12 (a) A constant (dc) signal and (b) its Fourier spectrum.

$x(t)$ is a dc signal that has a single frequency $\omega = 0$ (dc).

Fourier Transform: The $\delta(t)$ signal

EXAMPLE 7.5 Inverse Fourier Transform of a Shifted Dirac Delta Function

Find the inverse Fourier transform of $\delta(\omega - \omega_0)$.

Using the sampling property of the impulse function, we obtain

$$\mathcal{F}^{-1}[\delta(\omega - \omega_0)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t}$$

Therefore,

$$\frac{1}{2\pi} e^{j\omega_0 t} \Longleftrightarrow \delta(\omega - \omega_0) \quad \text{and} \quad e^{j\omega_0 t} \Longleftrightarrow 2\pi \delta(\omega - \omega_0) \quad (7.21)$$

$$e^{-j\omega_0 t} \Longleftrightarrow 2\pi \delta(\omega + \omega_0)$$

This result shows that the spectrum of an everlasting exponential $e^{j\omega_0 t}$ is a single impulse at $\omega = \omega_0$.

Fourier Transform: The Cosine signal

EXAMPLE 7.6 Fourier Transform of a Sinusoid

Find the Fourier transform of the everlasting sinusoid $\cos \omega_0 t$ (Fig. 7.13a).

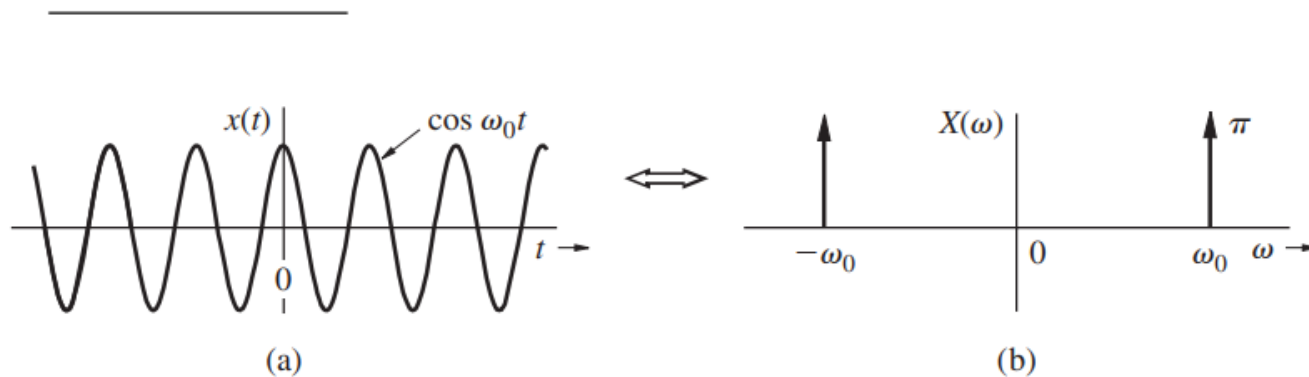


Figure 7.13 (a) A cosine signal and (b) its Fourier spectrum.

Recall Euler's formula

$$\cos \omega_0 t = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$$

Applying Eq. (7.21), we obtain

$$\cos \omega_0 t \longleftrightarrow \pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

An everlasting sinusoid $\cos \omega_0 t$ in the time domain is represented by two impulses at $\pm \omega_0$

$$e^{-j\omega_0 t} \longleftrightarrow 2\pi \delta(\omega + \omega_0)$$



Fourier Transform: The Sine signal

Find FT of $x(t) = \sin(\omega_0 t)$

$$x(t) = \sin(\omega_0 t)$$

$$\sin(\omega_0 t) = \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t})$$

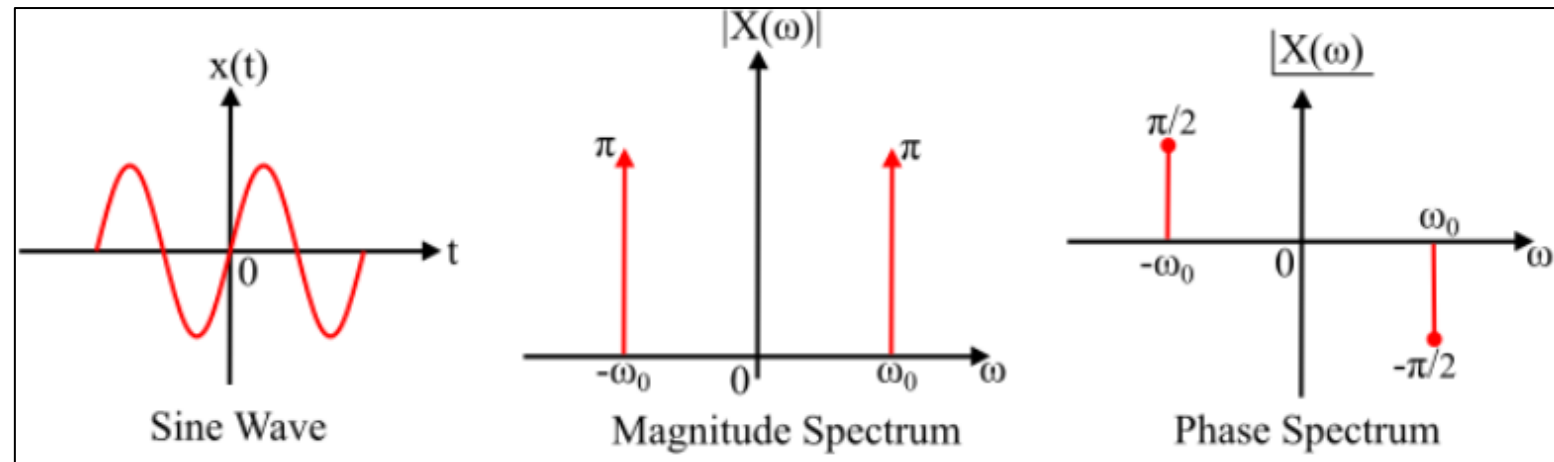
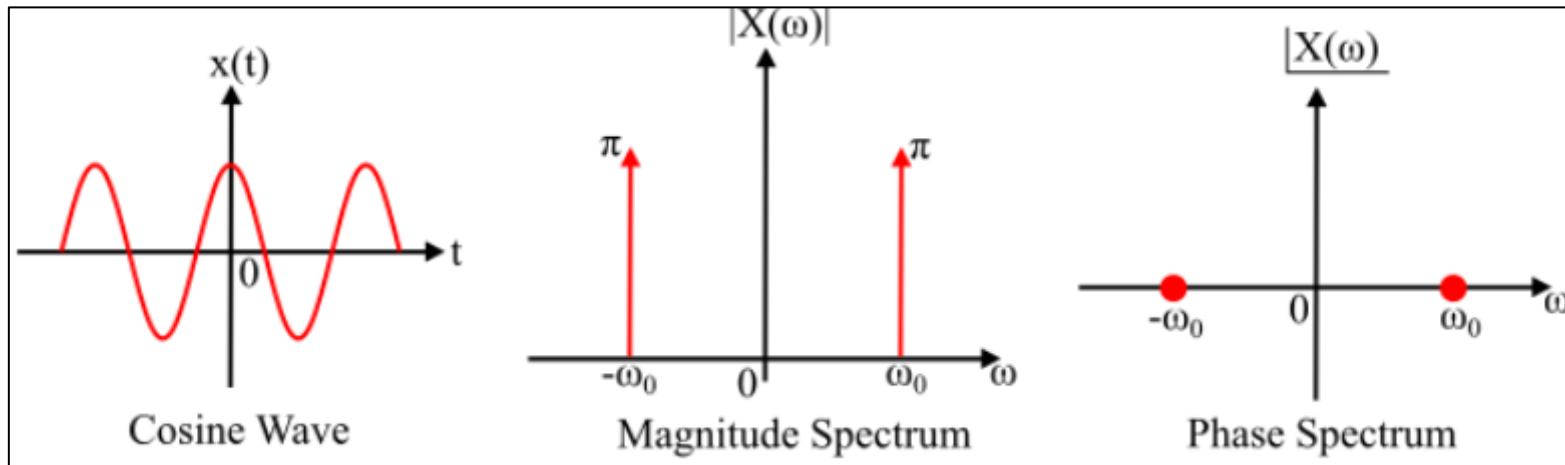
$$X(\omega) = \frac{1}{2j} [\mathcal{F}\{e^{j\omega_0 t}\} - \mathcal{F}\{e^{-j\omega_0 t}\}]$$

$$\mathcal{F}\{e^{j\omega_0 t}\} = 2\pi \delta(\omega - \omega_0), \quad \mathcal{F}\{e^{-j\omega_0 t}\} = 2\pi \delta(\omega + \omega_0)$$

$$X(\omega) = \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

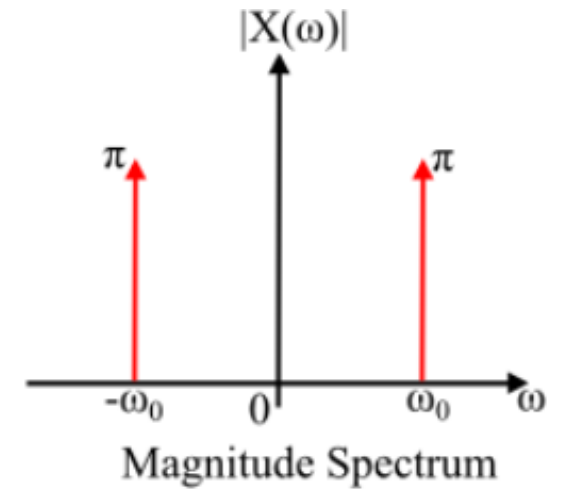
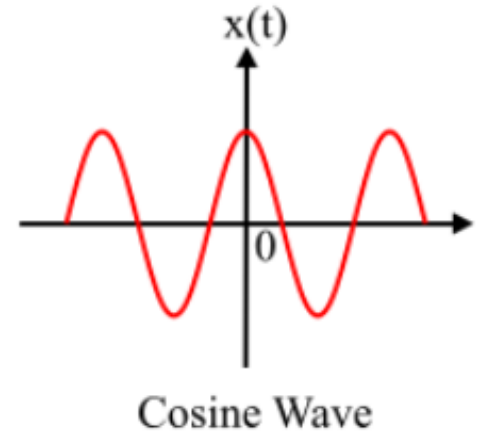
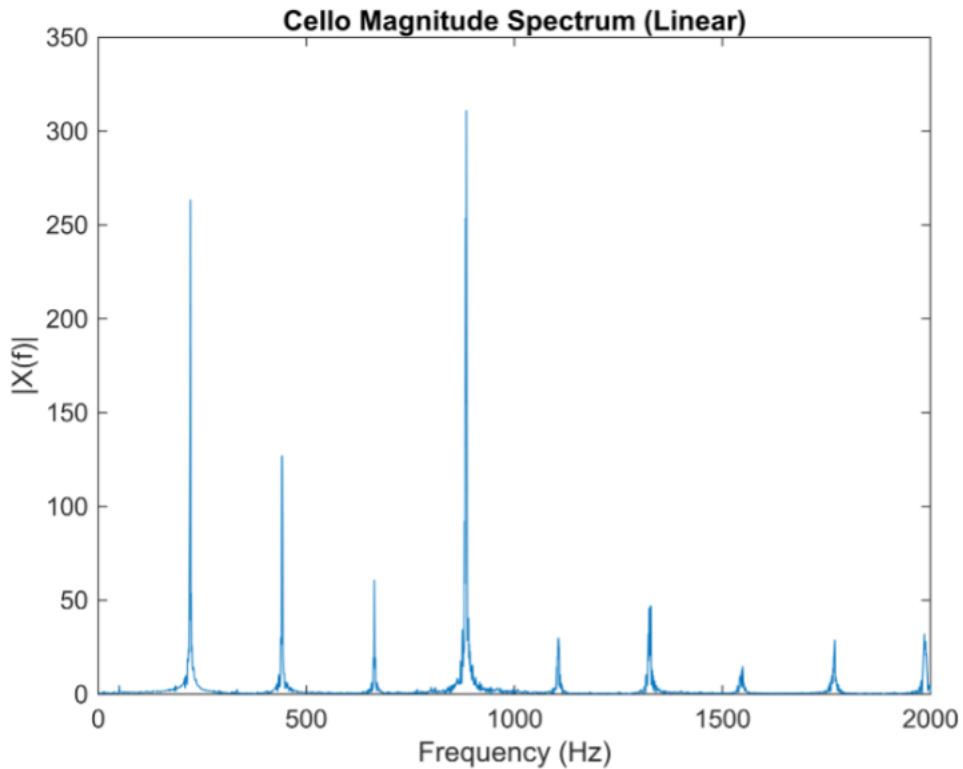
Fourier Transform: The Sinusoid signal

Find FT of $\cos(\omega_0 t)$ vs. $\sin(\omega_0 t)$



Fourier Transform:

SnS Project – Frequency Analysis of Cello and Trombone using FFT

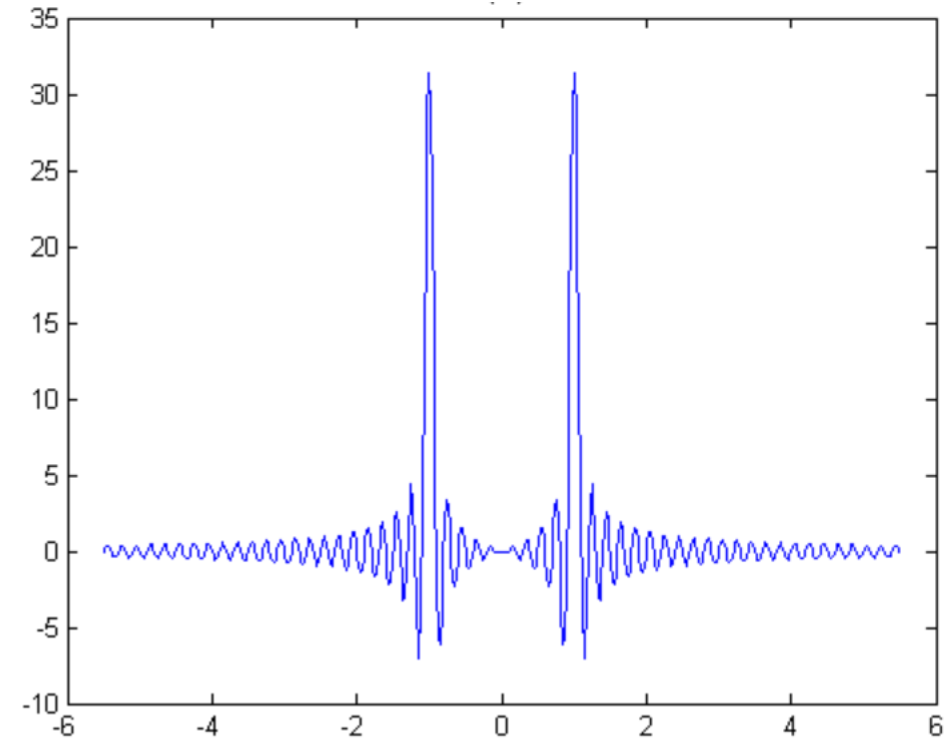
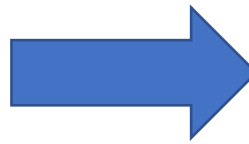
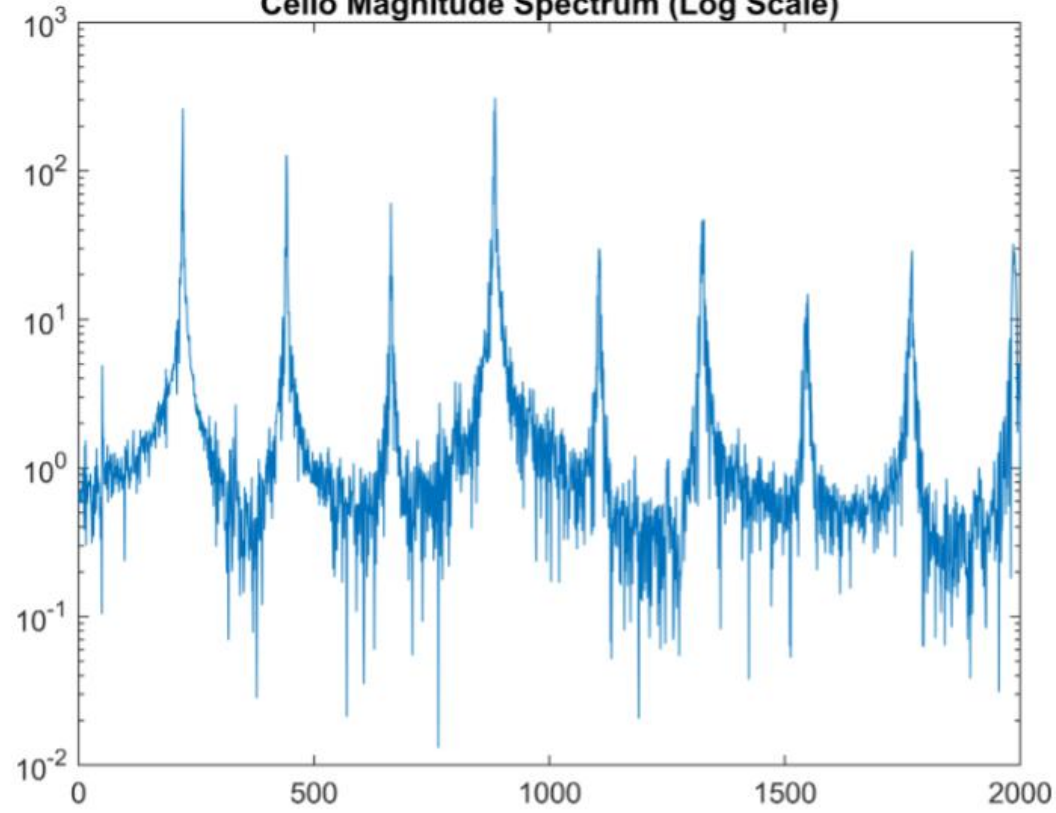


*Almost there – not
100% yet!*

Fourier Transform:

SnS Project – Frequency Analysis of Cello and Trombone using FFT

Cello Magnitude Spectrum (Log Scale)



Quiz4

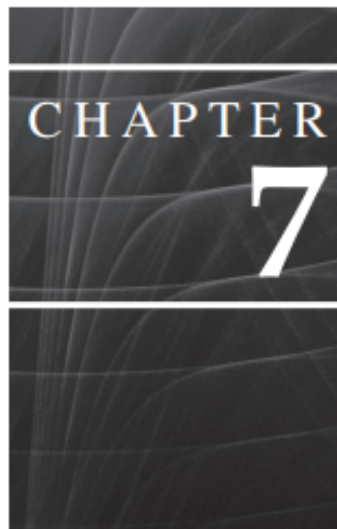
End of Session # 01



Learning outcome of Wk#13 session:

Frequency Domain Analysis using Fourier Transform(Chapter 07)

- CT Signal Representation by FT
- FT of basic signals
- FT properties
- FT of LTIC system



CONTINUOUS-TIME SIGNAL ANALYSIS: THE FOURIER TRANSFORM

Fourier Transform:

FT

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

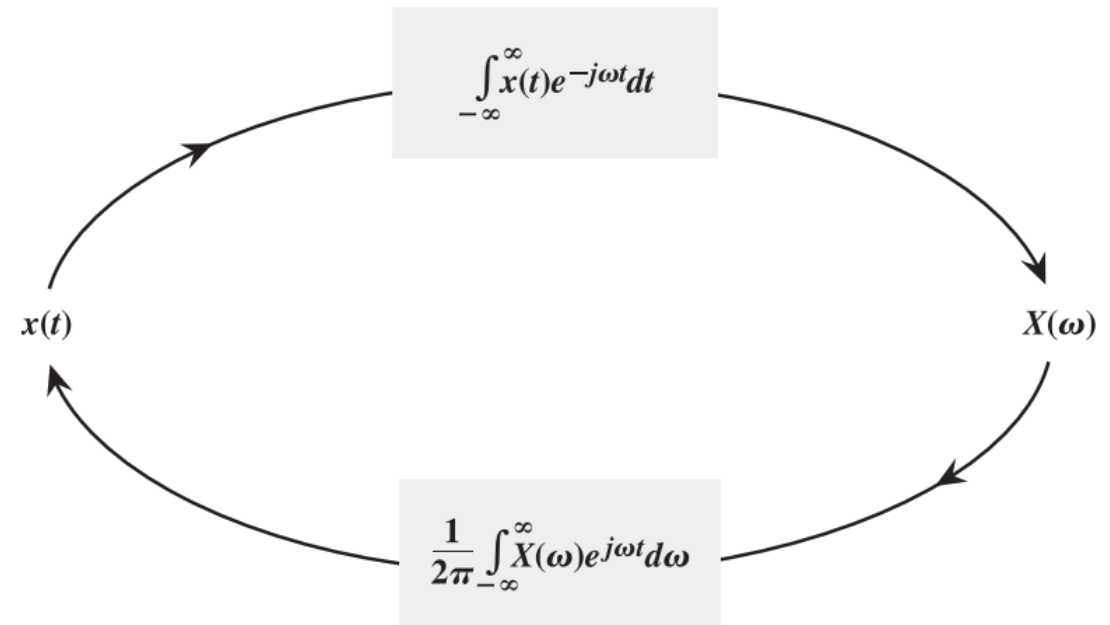
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FT pair notation

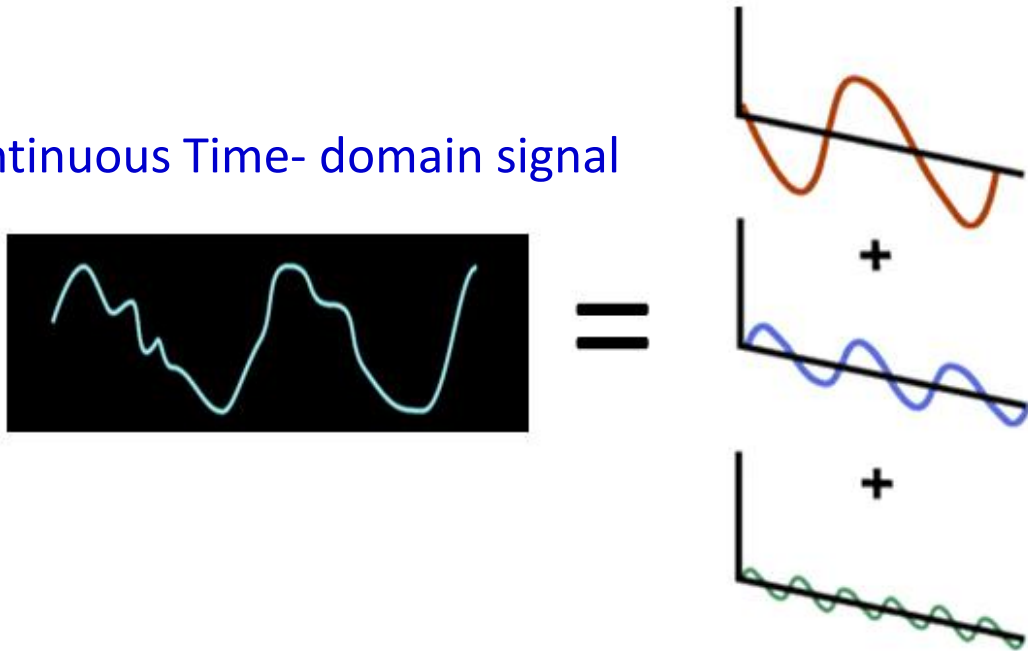
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$$x(t) \Longleftrightarrow X(\omega)$$

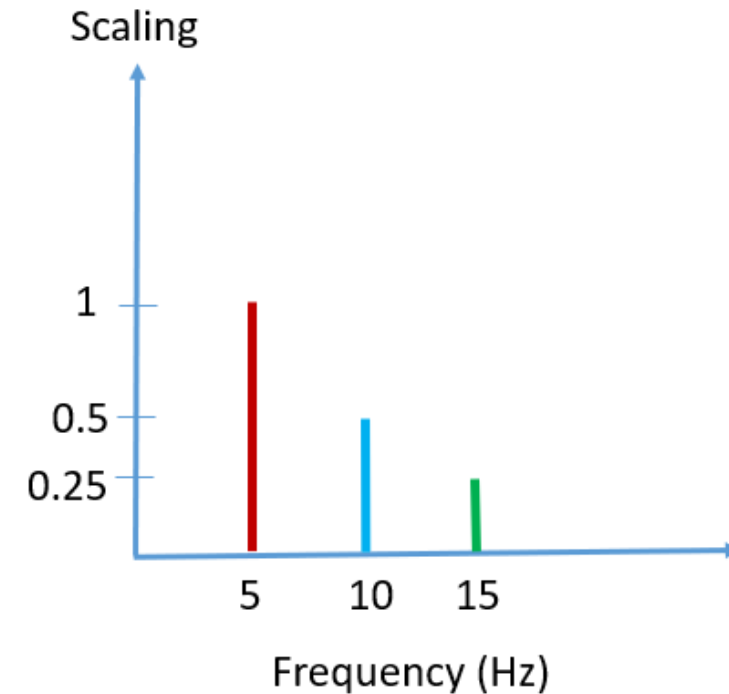


Fourier Transform:

Continuous Time- domain signal

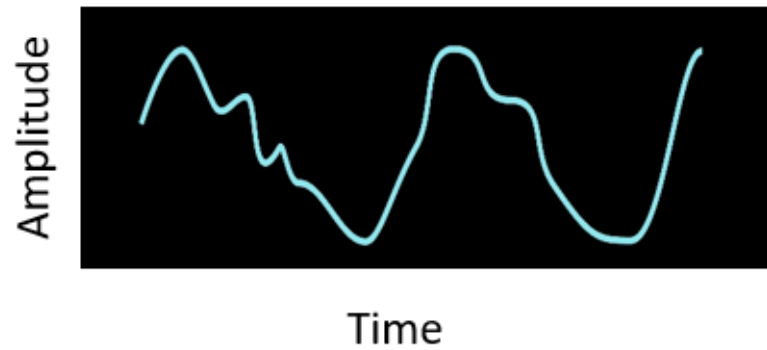


Fourier Series – decompose CT time domain signals into sine and cosine time series. Important insight: frequencies of sinusoids and their magnitude



Fourier Transform – frequency domain plot where each impulse represents a sinusoid

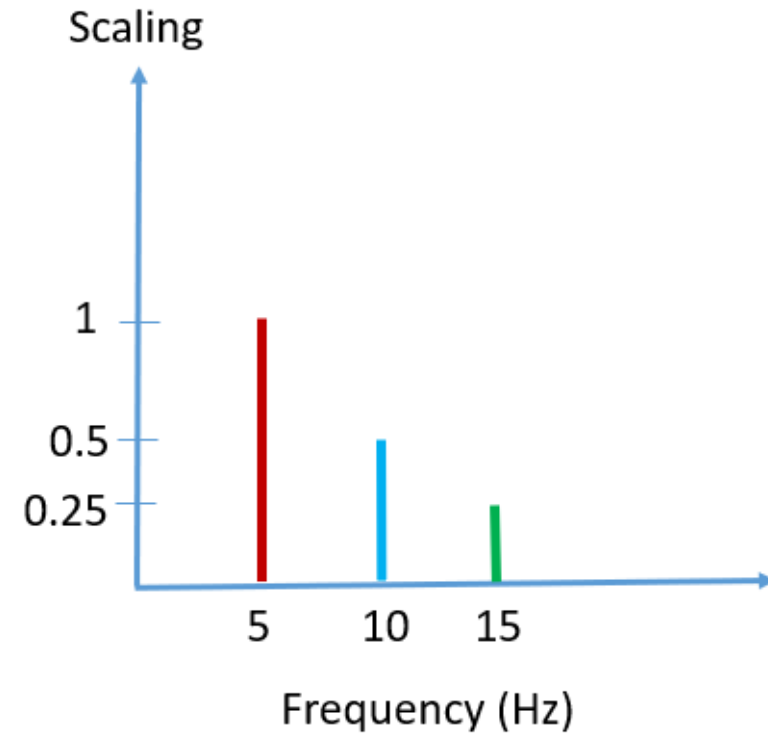
Fourier Transform:



Fourier Transform



Inverse
Fourier Transform



Fourier Transform:

FT Pairs

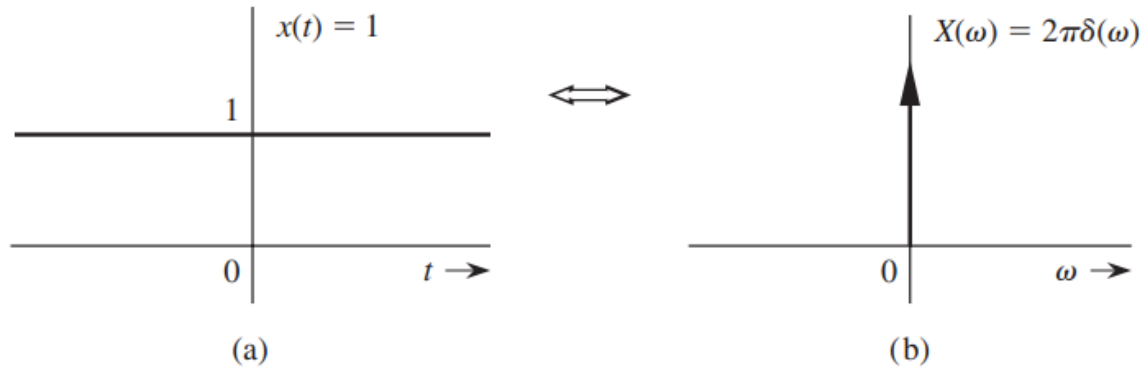
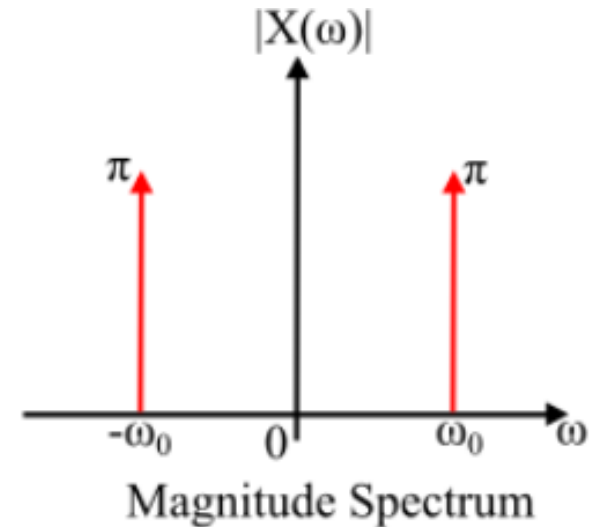
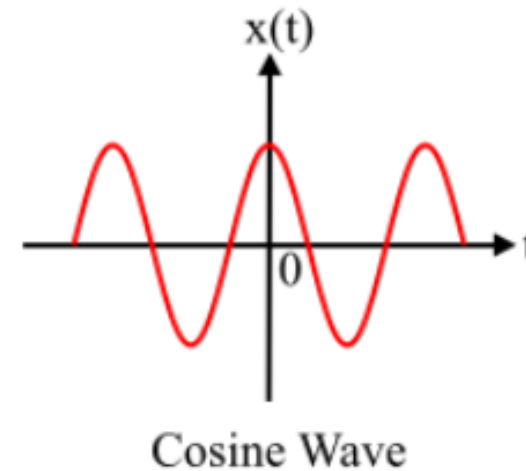
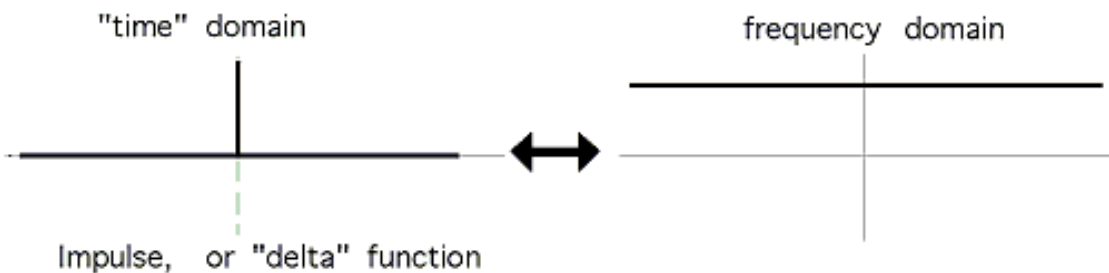
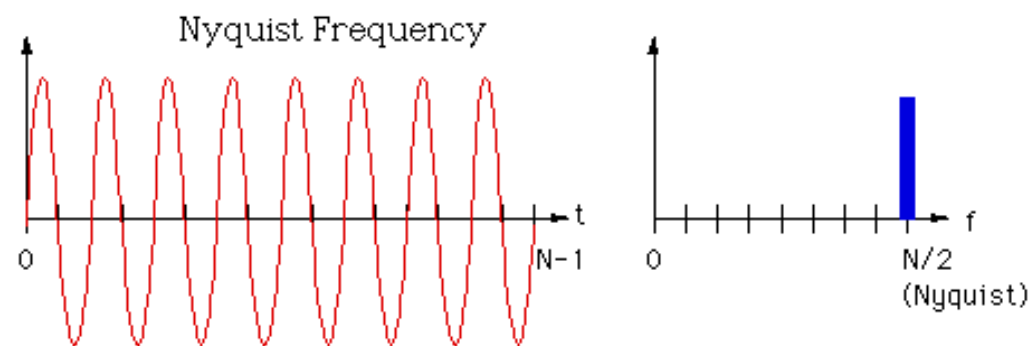
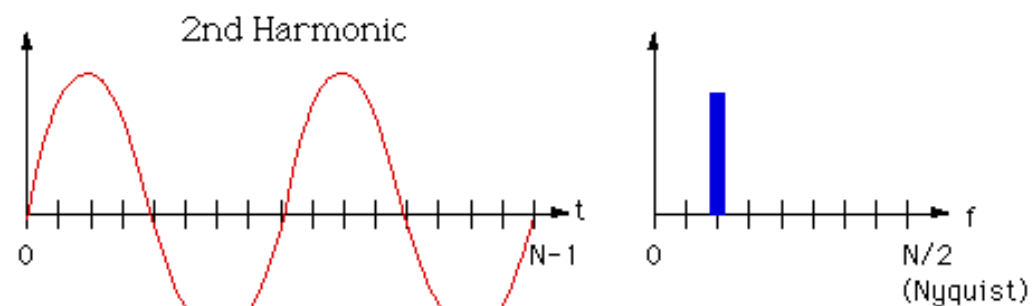
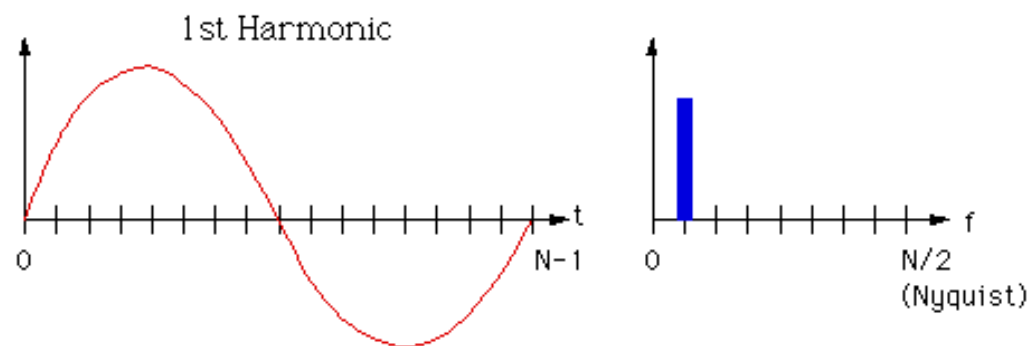
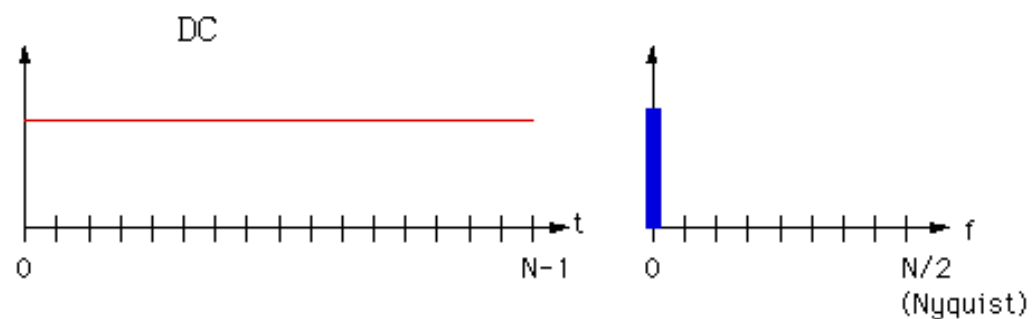


Figure 7.12 (a) A constant (dc) signal and (b) its Fourier spectrum.



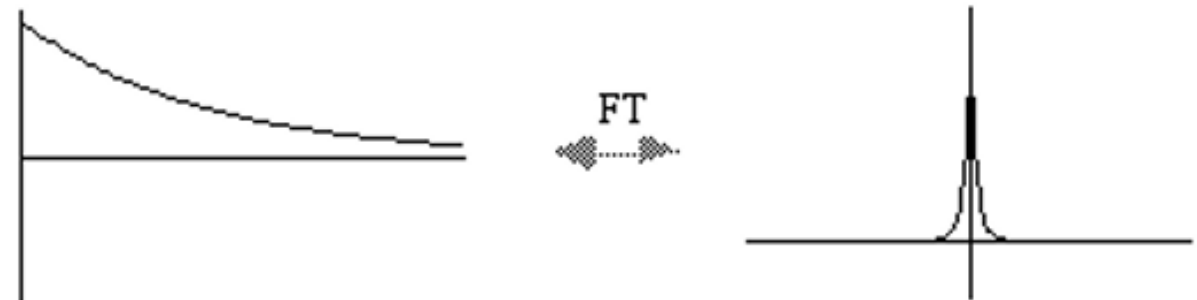
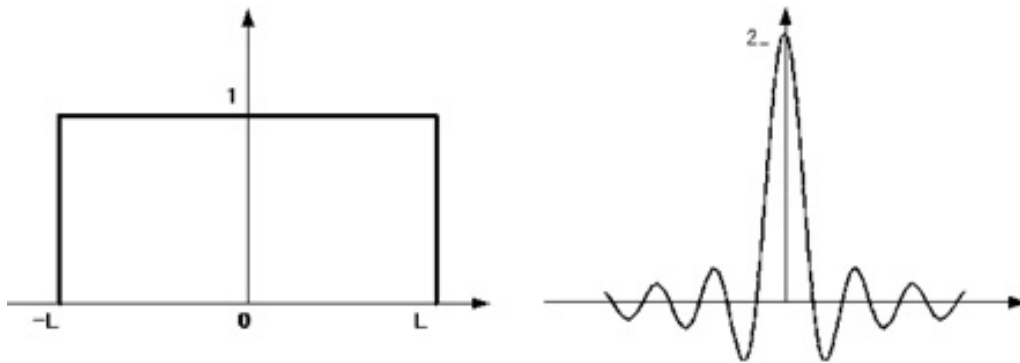
Fourier Transform:

FT Pairs



Fourier Transform:

FT Pairs



Fourier Transform: The Square signal

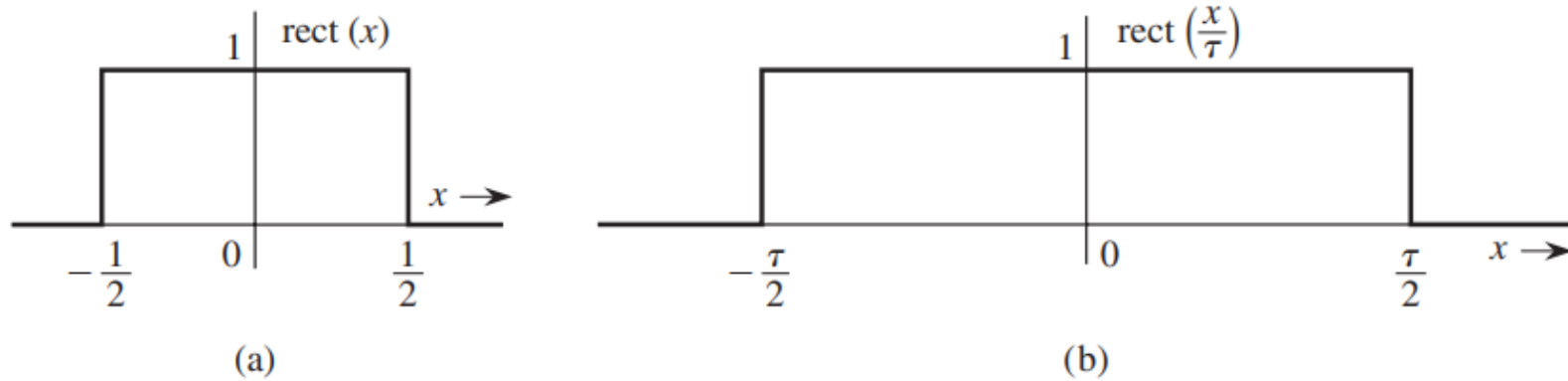


Figure 7.7 A gate pulse.

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\text{rect}(x) = \begin{cases} 0 & |x| > \frac{1}{2} \\ \frac{1}{2} & |x| = \frac{1}{2} \\ 1 & |x| < \frac{1}{2} \end{cases}$$



Fourier Transform: The Square signal

EXAMPLE 7.2 Fourier Transform of a Rectangular Pulse

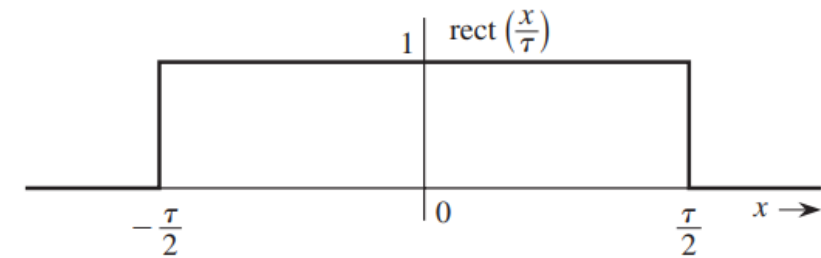
Find the Fourier transform of $x(t) = \text{rect}(t/\tau)$ (Fig. 7.10a).

$$X(\omega) = \int_{-\infty}^{\infty} \text{rect}\left(\frac{t}{\tau}\right) e^{-j\omega t} dt$$

Since $\text{rect}(t/\tau) = 1$ for $|t| < \tau/2$, and since it is zero for $|t| > \tau/2$,

$$\begin{aligned} X(\omega) &= \int_{-\tau/2}^{\tau/2} e^{-j\omega t} dt \\ &= -\frac{1}{j\omega} (e^{-j\omega\tau/2} - e^{j\omega\tau/2}) = \frac{2 \sin\left(\frac{\omega\tau}{2}\right)}{\omega} \\ &= \tau \frac{\sin\left(\frac{\omega\tau}{2}\right)}{\left(\frac{\omega\tau}{2}\right)} = \tau \text{sinc}\left(\frac{\omega\tau}{2}\right) \end{aligned}$$

$$\text{sinc}(x) = \frac{\sin \pi x}{\pi x}$$

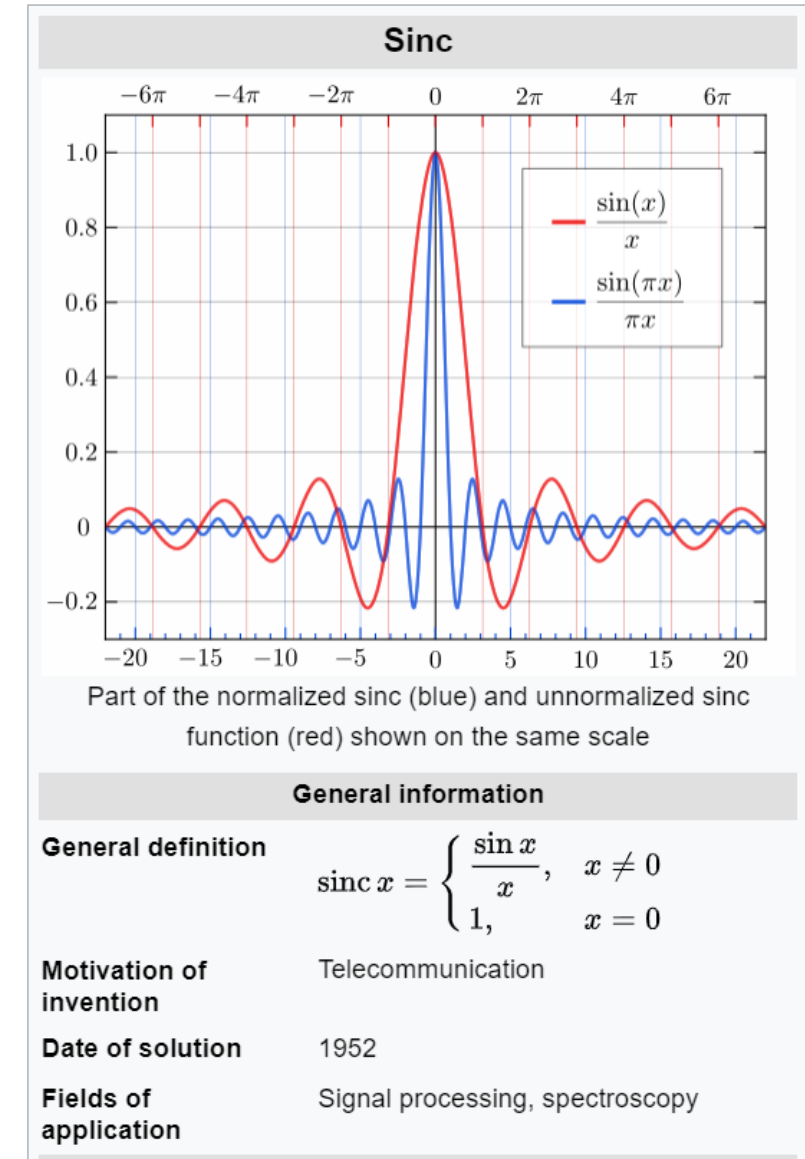


$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

The Sinc (x) signal:

The function has also been called the **cardinal sine** or **sine cardinal** function.^{[3][4]} The term *sinc* /'sɪŋk/ was introduced by [Philip M. Woodward](#) in his 1952 article "Information theory and inverse probability in telecommunication", in which he said that the function "occurs so often in Fourier analysis and its applications that it does seem to merit some notation of its own",^[5] and his 1953 book *Probability and Information Theory, with Applications to Radar*.^{[6][7]} The function itself was first mathematically derived in this form by [Lord Rayleigh](#) in his expression ([Rayleigh's formula](#)) for the zeroth-order spherical [Bessel function](#) of the first kind.

[Sinc function - Wikipedia](#)



The Sinc (x) signal:

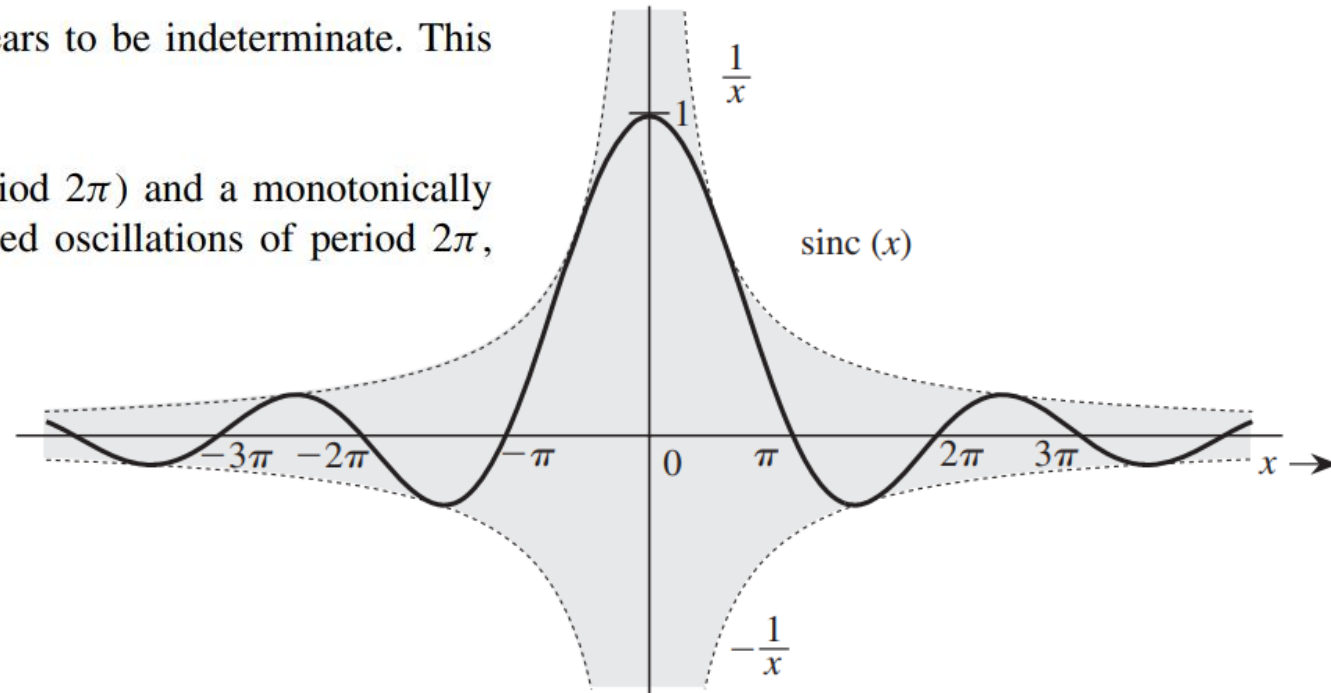
INTERPOLATION FUNCTION SINC (x)

The function $\sin x/x$ is the “sine over argument” function denoted by $\text{sinc}(x)$.[†] This function plays an important role in signal processing. It is also known as the *filtering or interpolating function*. We define

$$\text{sinc}(x) = \frac{\sin x}{x} \quad (7.18)$$

Inspection of Eq. (7.18) shows the following:

1. $\text{sinc}(x)$ is an even function of x .
2. $\text{sinc}(x) = 0$ when $\sin x = 0$ except at $x = 0$, where it appears to be indeterminate. This means that $\text{sinc} x = 0$ for $x = \pm\pi, \pm2\pi, \pm3\pi, \dots$
3. Using L'Hôpital's rule, we find $\text{sinc}(0) = 1$.
4. $\text{sinc}(x)$ is the product of an oscillating signal $\sin x$ (of period 2π) and a monotonically decreasing function $1/x$. Therefore, $\text{sinc}(x)$ exhibits damped oscillations of period 2π , with amplitude decreasing continuously as $1/x$.





Fourier Transform: The Square signal

EXAMPLE 7.2 Fourier Transform of a Rectangular Pulse

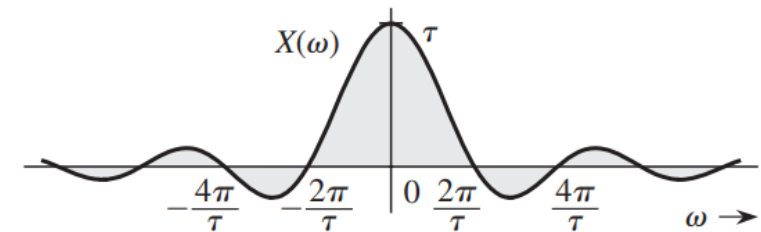
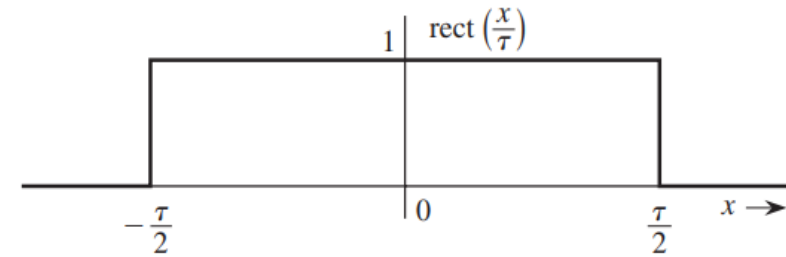
Find the Fourier transform of $x(t) = \text{rect}(t/\tau)$ (Fig. 7.10a).

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$\text{sinc}(x) = \frac{\sin \pi x}{\pi x}$



$$\text{sinc}(0) = 1$$

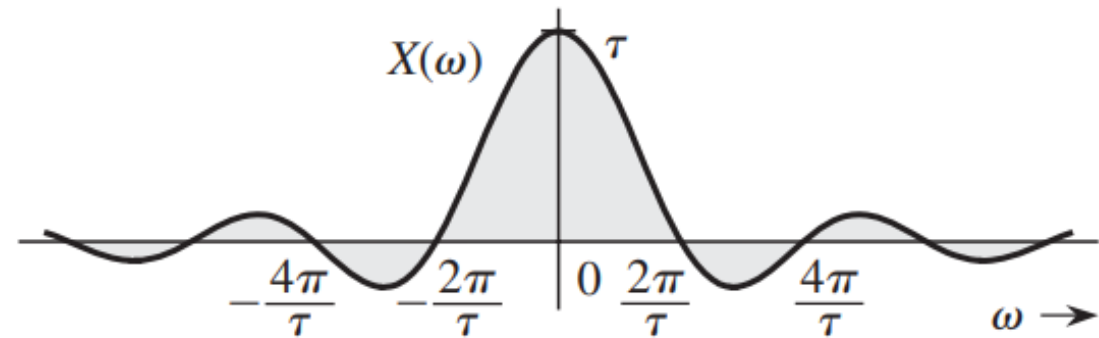
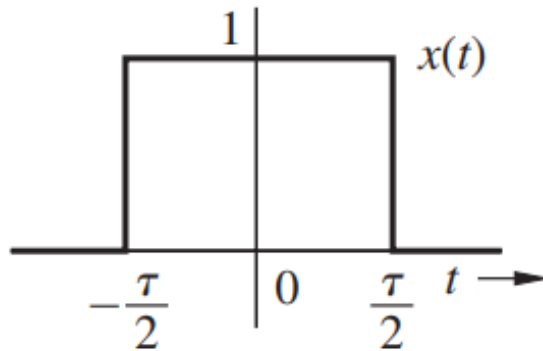
As, $\text{sinc}(x) = 0$ when $x = \pm n\pi$

$\text{sinc}(\omega\tau/2) = 0$ when $\omega\tau/2 = \pm n\pi$;

$$\omega = \pm 2n\pi/\tau, (n = 1, 2, 3, \dots),$$

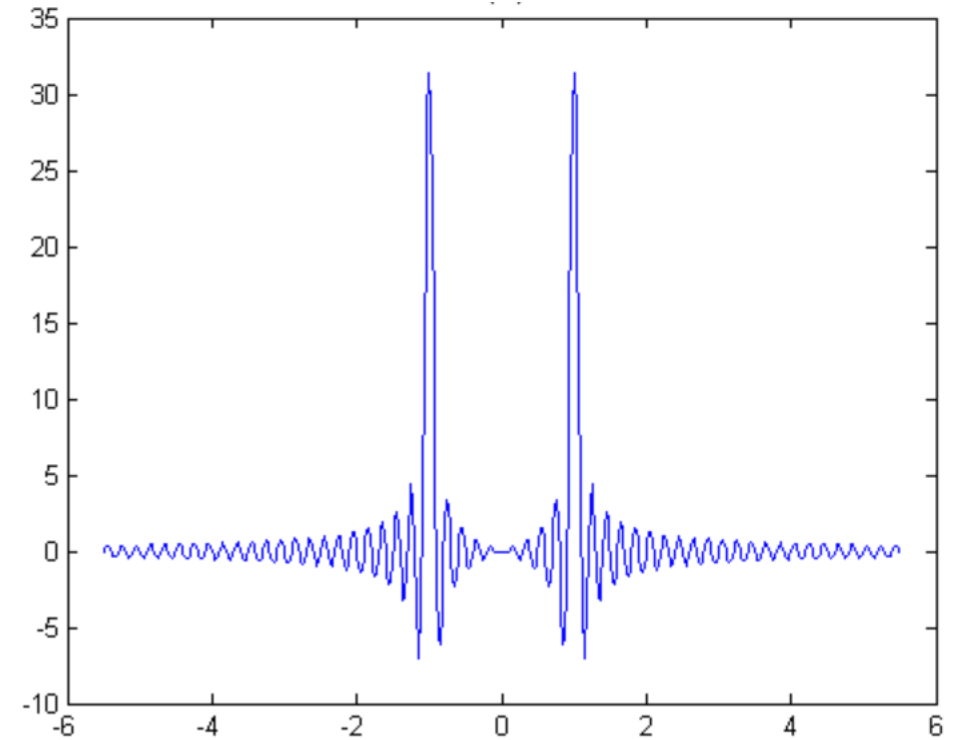
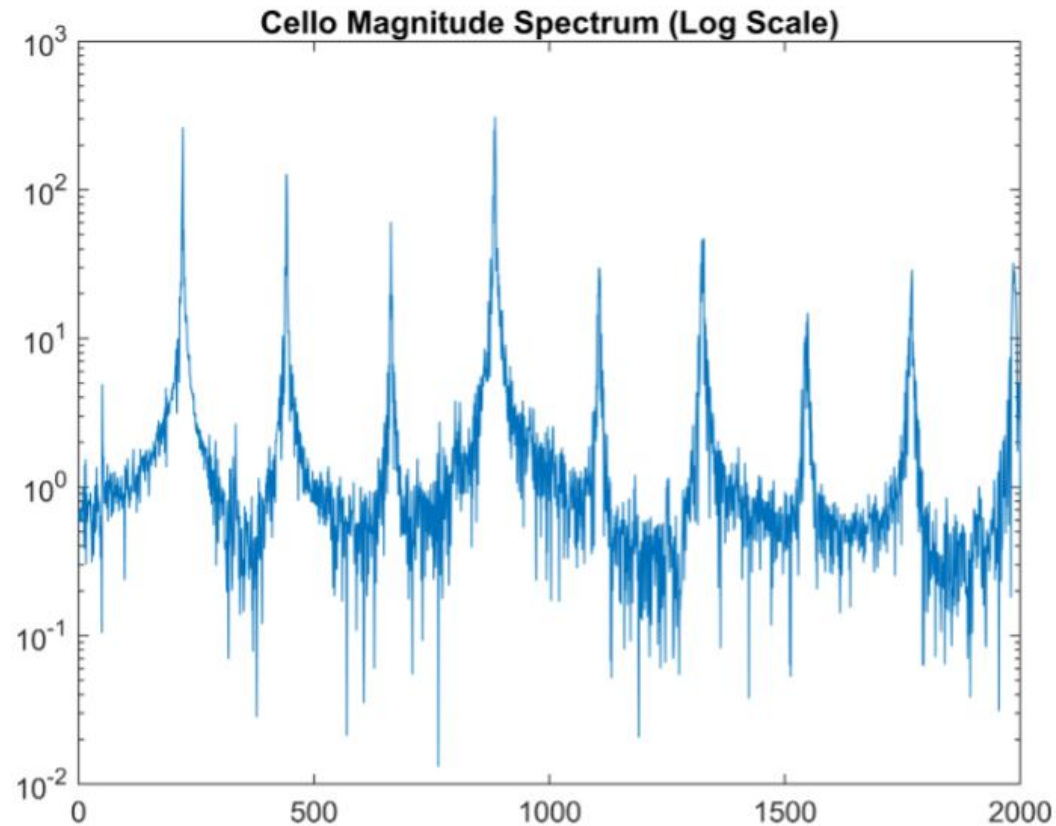
Fourier Transform: The Square signal

$$\text{rect}\left(\frac{t}{\tau}\right) \Longleftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$



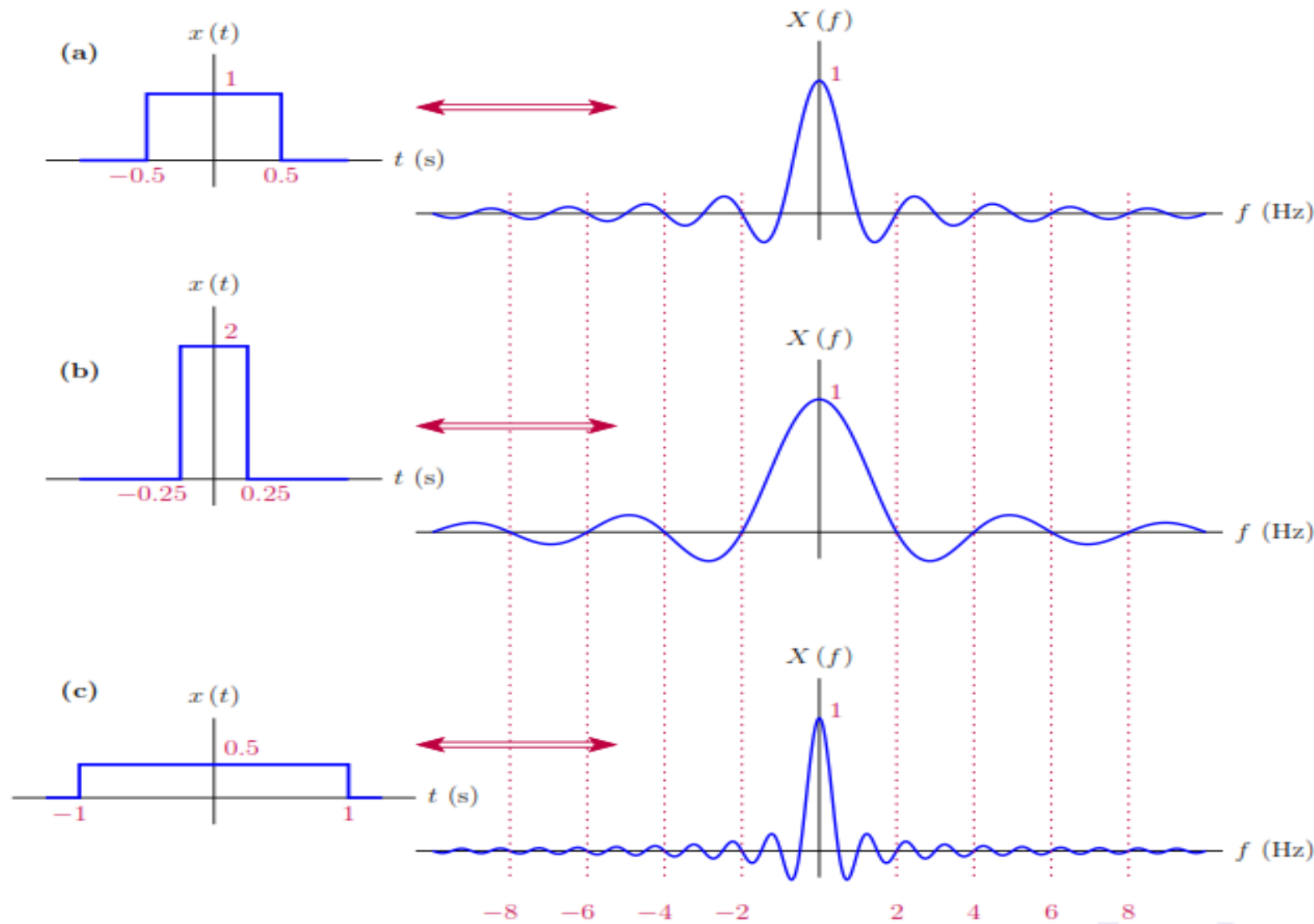
Fourier Transform:

SnS Project – Frequency Analysis of Cello and Trombone using FFT



So, each spike is a $\text{sinc}(x)$ signal – a windowed cosine signal

Fourier Transform: The Square signal



A narrow signal in the time domain is wider in the frequency domain --- FT Duality property

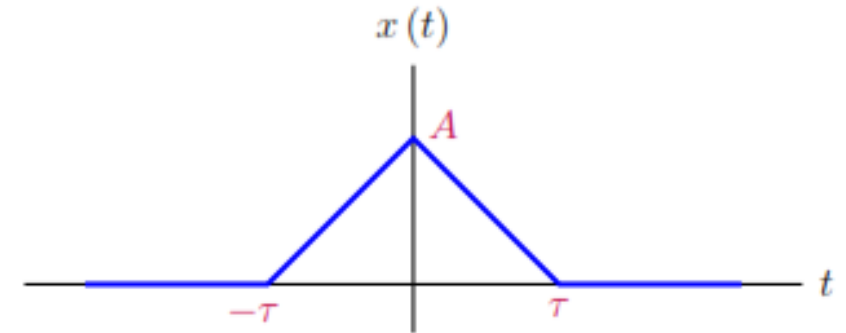
Fourier Transform: The Triangular signal

Fourier transform of a triangular pulse

Find the Fourier transform of the triangular pulse signal given by

$$x(t) = A \Lambda\left(\frac{t}{\tau}\right) = \begin{cases} A + At/\tau, & -\tau < t < 0 \\ A - At/\tau, & 0 \leq t < \tau \\ 0, & |t| \geq \tau \end{cases}$$

where $\Lambda(t)$ is the unit-triangle function.



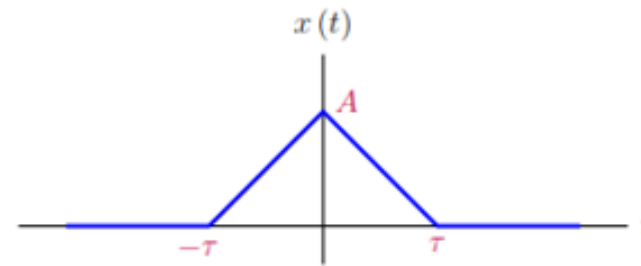
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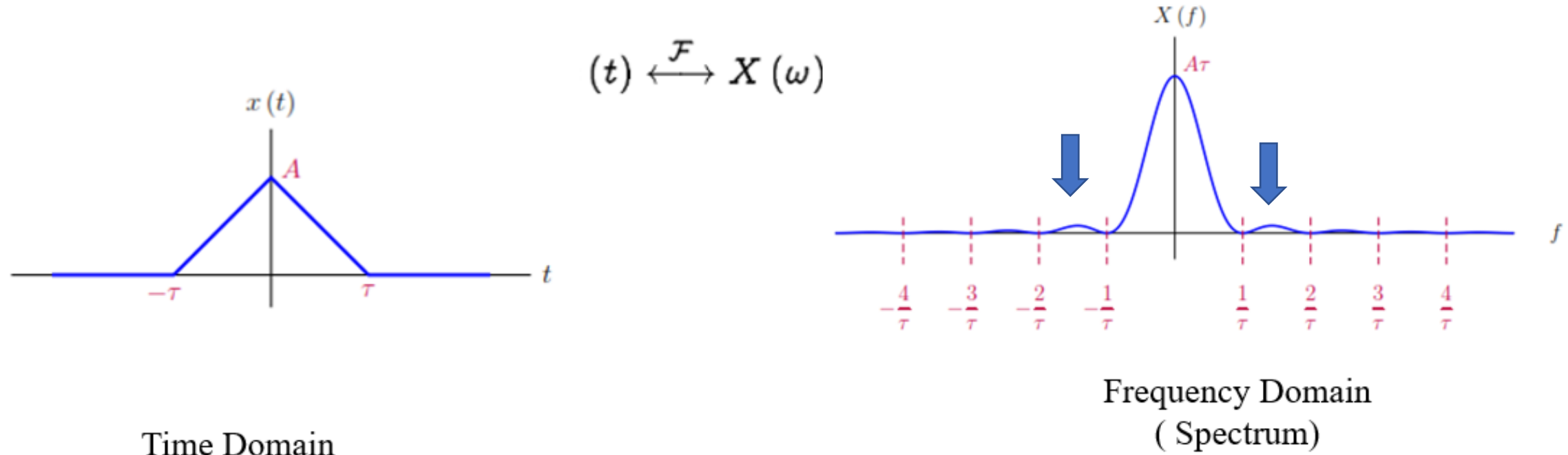
Solution:

$$X(\omega) = \int_{-\tau}^0 \left(A + \frac{At}{\tau}\right) e^{-j\omega t} dt + \int_0^{\tau} \left(A - \frac{At}{\tau}\right) e^{-j\omega t} dt = \frac{4A}{\omega^2 \tau} \sin^2\left(\frac{\omega \tau}{2}\right)$$

Use the sinc function:

$$X(\omega) = A\tau \operatorname{sinc}^2\left(\frac{\omega \tau}{2\pi}\right)$$

Fourier Transform: The Triangular signal



Fourier Transform: The Exponential signal

EXAMPLE 7.1 Fourier Transform of a Causal Exponential

Find the Fourier transform of $e^{-at}u(t)$.

By definition [Eq. (7.9)],

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j\omega t} dt = \int_0^{\infty} e^{-(a+j\omega)t} dt = \frac{-1}{a+j\omega} e^{-(a+j\omega)t} \Big|_0^{\infty}$$

But $|e^{-j\omega t}| = 1$. Therefore, as $t \rightarrow \infty$, $e^{-(a+j\omega)t} = e^{-at}e^{-j\omega t} = \infty$ if $a < 0$, but it is equal to 0 if $a > 0$. Therefore,

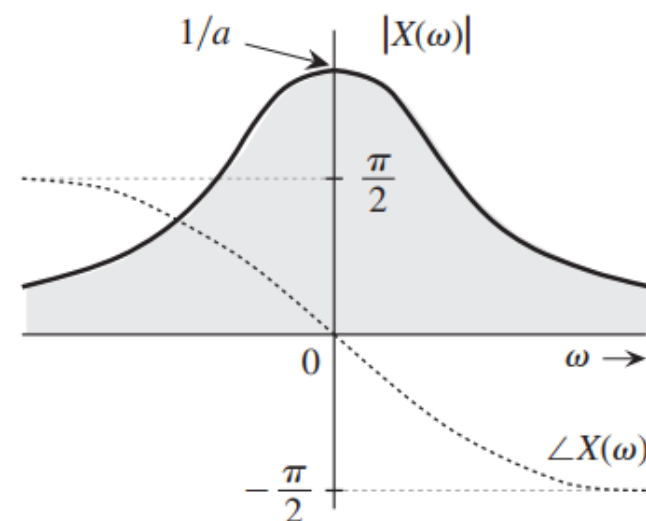
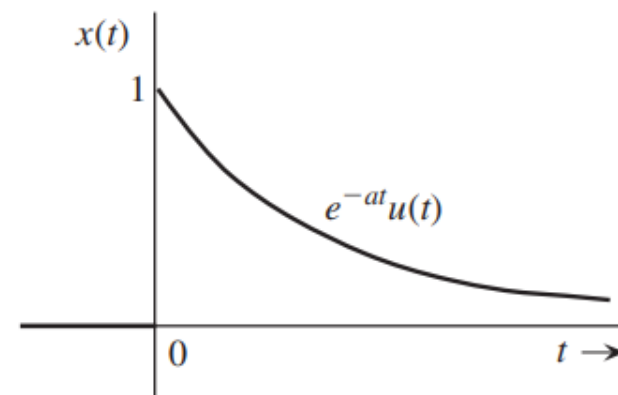
$$X(\omega) = \frac{1}{a+j\omega} \quad a > 0$$

Expressing $a+j\omega$ in the polar form as $\sqrt{a^2+\omega^2}e^{j\tan^{-1}(\omega/a)}$, we obtain

$$X(\omega) = \frac{1}{\sqrt{a^2+\omega^2}} e^{-j\tan^{-1}(\omega/a)}$$

Therefore,

$$|X(\omega)| = \frac{1}{\sqrt{a^2+\omega^2}} \quad \text{and} \quad \angle X(\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$





Fourier Transform: The Exponential signal

Exponentially decaying signal

$$x(t) = e^{-at}u(t), \quad a > 0$$

$$X(\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt = \int_0^{\infty} e^{-(a+j\omega)t} dt$$

This converges only if $a > 0$.

$$X(\omega) = \frac{1}{a + j\omega}$$

$$X(\omega) = \frac{1}{a + j\omega}, \quad a > 0$$

$$x(t) = e^{at}u(-t), \quad a > 0$$

$$X(\omega) = \int_{-\infty}^0 e^{at} e^{-j\omega t} dt = \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

This converges because $t \rightarrow -\infty$.

$$X(\omega) = \frac{1}{a - j\omega}$$

$$X(\omega) = \frac{1}{a - j\omega}, \quad a > 0$$

Exponentially increasing signal

$$x(t) = e^{at}u(t), \quad a > 0$$

$$\int_0^{\infty} e^{at} dt \rightarrow \infty$$

Fourier Transform does NOT exist



Fourier Transform: Not possible for all signals!

Conditions for the Existence of the Fourier Transform:

For a signal $x(t)$, the Fourier Transform exists if the convergence conditions are satisfied

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

To ensure the FT integral (shown below) converges absolutely, the signal must have a finite area

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Fourier Transform:

TABLE 7.1 Select Fourier Transform Pairs

No.	$x(t)$	$X(\omega)$	
1	$e^{-at}u(t)$	$\frac{1}{a + j\omega}$	$a > 0$
2	$e^{at}u(-t)$	$\frac{1}{a - j\omega}$	$a > 0$
3	$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$	$a > 0$
4	$te^{-at}u(t)$	$\frac{1}{(a + j\omega)^2}$	$a > 0$
5	$t^n e^{-at}u(t)$	$\frac{n!}{(a + j\omega)^{n+1}}$	$a > 0$
6	$\delta(t)$	1	
7	1	$2\pi\delta(\omega)$	
8	$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	
9	$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
10	$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$	
11	$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$	



Fourier Transform:

TABLE 7.1 Select Fourier Transform Pairs

No.	$x(t)$	$X(\omega)$	
12	$\text{sgn } t$	$\frac{2}{j\omega}$	
13	$\cos \omega_0 t u(t)$	$\frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$	
14	$\sin \omega_0 t u(t)$	$\frac{\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + \frac{\omega_0}{\omega_0^2 - \omega^2}$	
15	$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$	$a > 0$
16	$e^{-at} \cos \omega_0 t u(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$	$a > 0$
17	$\text{rect}\left(\frac{t}{\tau}\right)$	$\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$	
18	$\frac{W}{\pi} \text{sinc}(Wt)$	$\text{rect}\left(\frac{\omega}{2W}\right)$	
19	$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \text{sinc}^2\left(\frac{\omega\tau}{4}\right)$	
20	$\frac{W}{2\pi} \text{sinc}^2\left(\frac{Wt}{2}\right)$	$\Delta\left(\frac{\omega}{2W}\right)$	
21	$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$	$\omega_0 = \frac{2\pi}{T}$
22	$e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi} e^{-\sigma^2\omega^2/2}$	

Fourier Transform Properties



Fourier Transform:

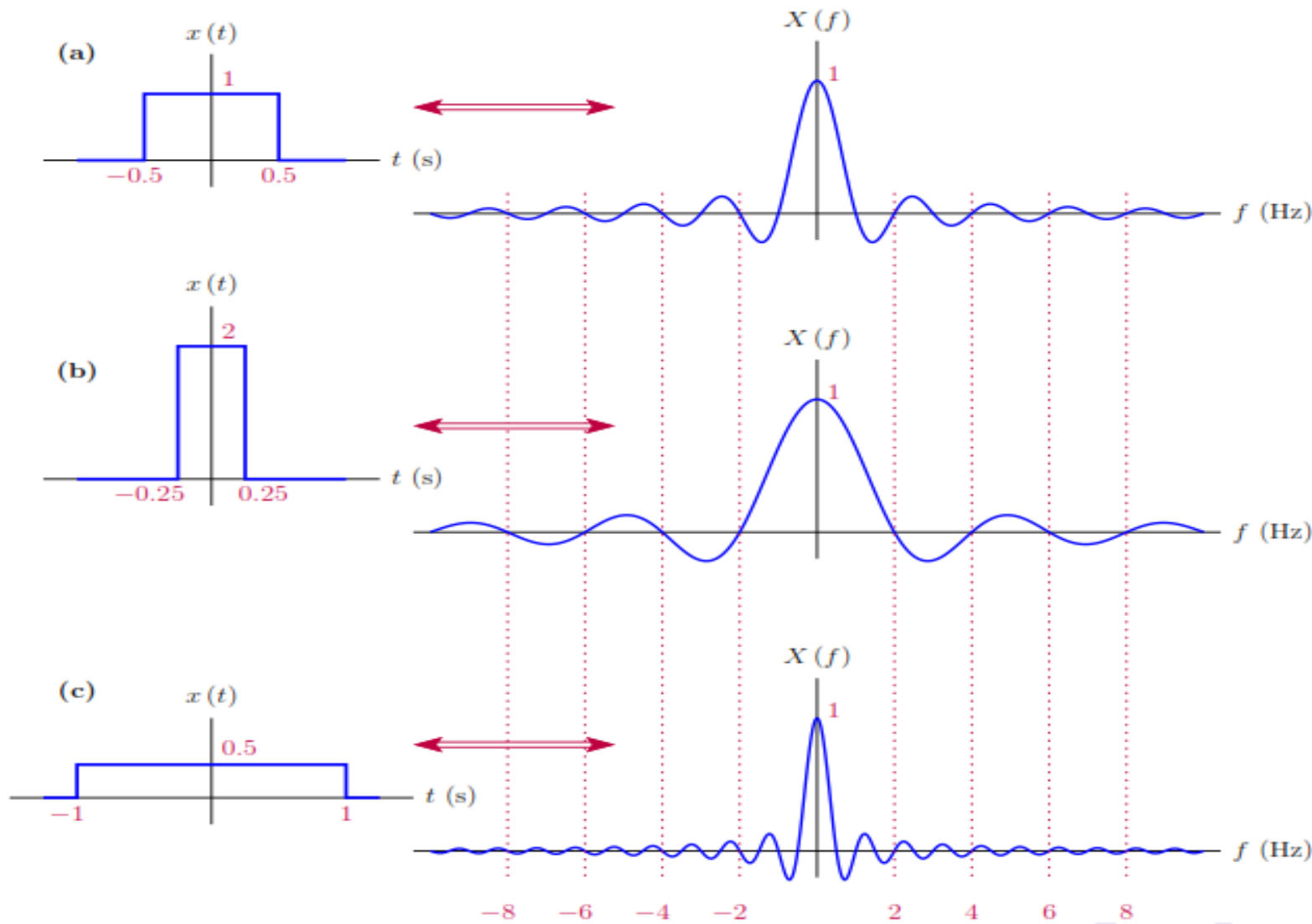
Properties of the FT

Read the book and poof all these properties- Important for quiz and final exam.

TABLE 7.2 Fourier Transform Properties

Operation	$x(t)$	$X(\omega)$
Scalar multiplication	$kx(t)$	$kX(\omega)$
Addition	$x_1(t) + x_2(t)$	$X_1(\omega) + X_2(\omega)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Scaling (a real)	$x(at)$	$\frac{1}{ a } X\left(\frac{\omega}{a}\right)$
Time shifting	$x(t - t_0)$	$X(\omega)e^{-j\omega t_0}$
Frequency shifting (ω_0 real)	$x(t)e^{j\omega_0 t}$	$X(\omega - \omega_0)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(\omega)X_2(\omega)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
Time differentiation	$\frac{d^n x(t)}{dt^n}$	$(j\omega)^n X(\omega)$
Time integration	$\int_{-\infty}^t x(u) du$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$

Fourier Transform:

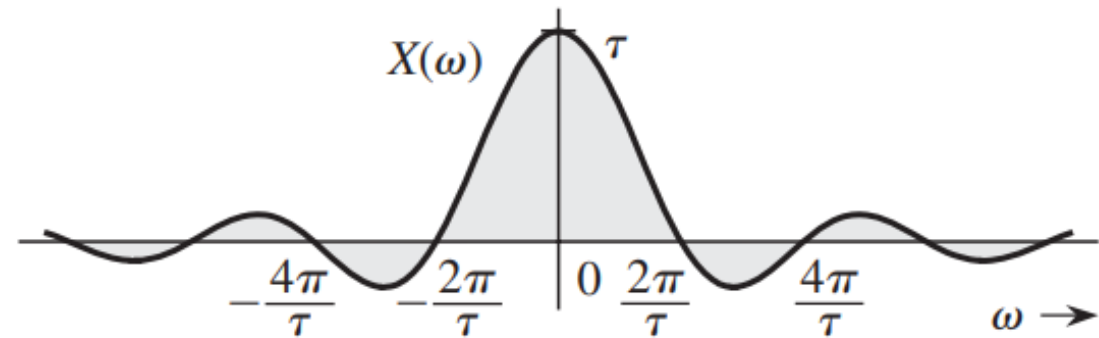
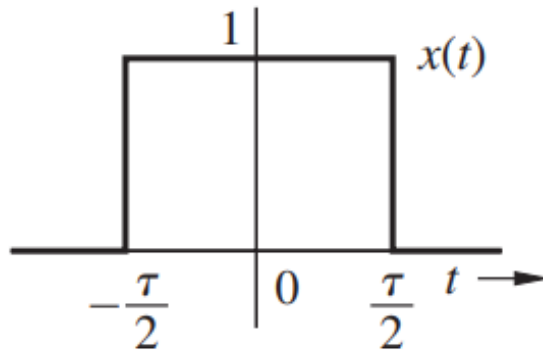


A narrow signal in the time domain is wider in the frequency domain --- Duality property of FT

Fourier Transform:

$$\text{rect}\left(\frac{t}{\tau}\right) \Longleftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

Homework4



Try the amplified, shifted, and extended version of the pulse, and by using the FT properties/table, find its FT and plot it!



Fourier Transform:

Transforms of even and odd signals:

Real valued signal with even symmetry

$$x(-t) = x(t) , \text{all } t \quad \text{implies that} \quad \text{Im} \{X(\omega)\} = 0 , \text{all } \omega$$

Real valued signal with odd symmetry

$$x(-t) = -x(t) , \text{all } t \quad \text{implies that} \quad \text{Re} \{X(\omega)\} = 0 , \text{all } \omega$$

Fourier Transform:

Transform of a two-sided exponential signal

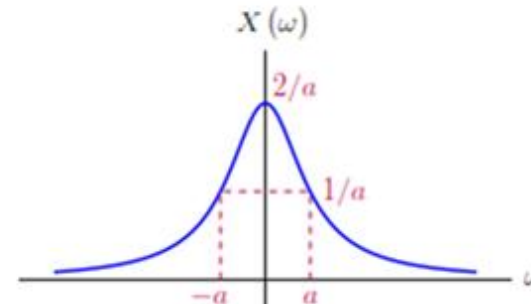
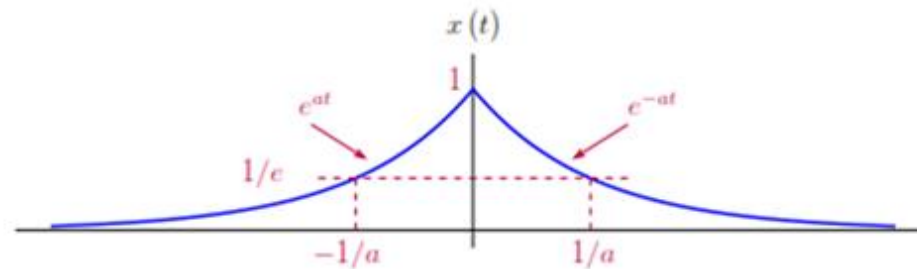
Elaborate on the symmetry properties of the Fourier transform of the exponential signal $x(t) = e^{-a|t|}$.

Solution:

The Fourier transform is

$$X(\omega) = \frac{2a}{a^2 + \omega^2}$$

$X(\omega)$ is real.



Real valued signal with even symmetry

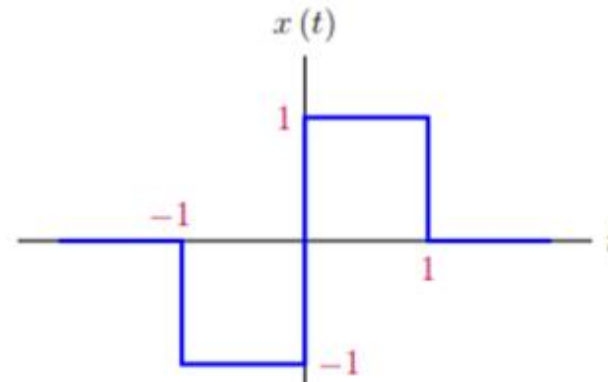
$x(-t) = x(t)$, all t implies that $\text{Im}\{X(\omega)\} = 0$, all ω

Fourier Transform:

Transform of a pulse with odd symmetry

Determine the Fourier transform of the signal

$$x(t) = \begin{cases} -1, & -1 < t < 0 \\ 1, & 0 < t < 1 \\ 0, & t < -1 \text{ or } t > 1 \end{cases}$$



Question: through $X(\omega)$ identify the Symmetry of the signal (Even/Odd)?

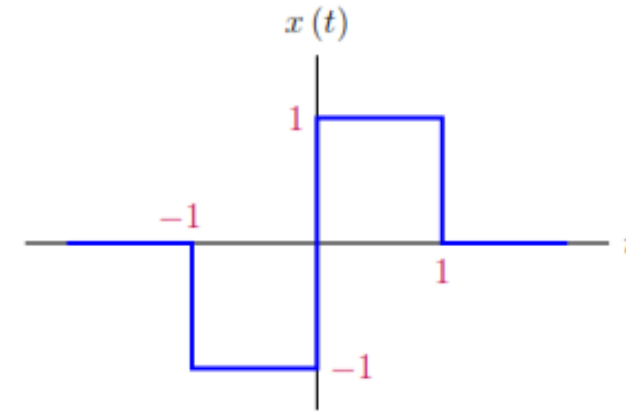
Fourier Transform:

Transform of a pulse with odd symmetry

Determine the Fourier transform of the signal

$$x(t) = \begin{cases} -1, & -1 < t < 0 \\ 1, & 0 < t < 1 \\ 0, & t < -1 \text{ or } t > 1 \end{cases}$$

and show that it is purely imaginary.



Solution:

Through direct use of the forward transform integral:

$$X(\omega) = \int_{-1}^0 (-1) e^{-j\omega t} dt + \int_0^1 (1) e^{-j\omega t} dt = \frac{j2}{\omega} [\cos(\omega) - 1]$$

$X(\omega)$ = Imaginary

Real valued signal with odd symmetry

$$x(-t) = -x(t), \text{ all } t \text{ implies that } \operatorname{Re}\{X(\omega)\} = 0, \text{ all } \omega$$

Fourier Transform:

Linearity

$$x(t) = \begin{cases} t e^{-3t}, & t \geq 0, \\ (2 - t) e^{2t}, & t < 0, \end{cases}$$

Linearity of the Fourier transform

For any two constants α_1 and α_2 :

$$\alpha_1 x_1(t) + \alpha_2 x_2(t) \xleftrightarrow{\mathcal{F}} \alpha_1 X_1(\omega) + \alpha_2 X_2(\omega)$$

$$X(\omega) = \underbrace{\int_0^{\infty} t e^{-3t} e^{-j\omega t} dt}_{X_1(\omega)} + \underbrace{\int_{-\infty}^0 (2 - t) e^{2t} e^{-j\omega t} dt}_{X_2(\omega)}.$$

$$X(\omega) = \frac{1}{(3 + j\omega)^2} + \frac{2}{2 - j\omega} + \frac{1}{(2 - j\omega)^2}.$$

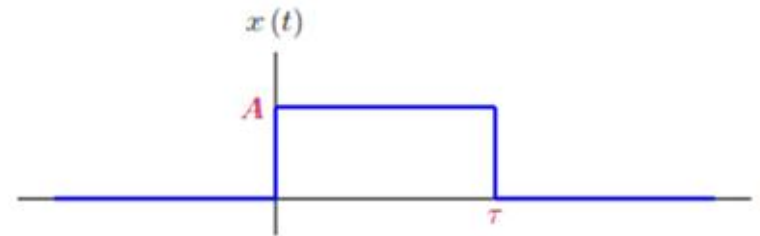
Properties of Fourier Transform:

Time shifting

Time shifting a rectangular pulse

Using the time shifting property, find the transform of the isolated rectangular pulse given by

$$x(t) = A \Pi \left(\frac{t - \tau/2}{\tau} \right)$$



Solution:

The transform of a rectangular pulse with amplitude A , width τ and center at $t = 0$ was found in Example 4.12:

$$A \Pi \left(\frac{t}{\tau} \right) \xleftrightarrow{\mathcal{F}} A\tau \operatorname{sinc} \left(\frac{\omega\tau}{2} \right)$$

Use time-shifting property:

$$A \Pi \left(\frac{t - \tau/2}{\tau} \right) \xleftrightarrow{\mathcal{F}} A\tau \operatorname{sinc} \left(\frac{\omega\tau}{2} \right) e^{-j\omega\tau/2}$$

Properties of Fourier Transform:

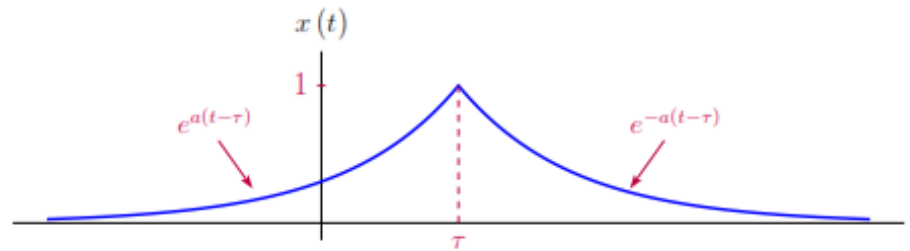
Time shifting

Time shifting a two-sided exponential signal

Determine the Fourier transform of the signal

$$x(t) = e^{-a|t-\tau|}$$

where $a > 0$.



Solution:

In Example 4.16 it was determined that

$$\mathcal{F}\{e^{-a|t|}\} = \frac{2a}{a^2 + \omega^2}$$

Use the time shifting property:

$$X(\omega) = \mathcal{F}\{e^{-a|t-\tau|}\} = \frac{2a e^{-j\omega\tau}}{a^2 + \omega^2}$$

Two-sided exponential	$x(t) = e^{-a t }$	$X(\omega) = \frac{2a}{a^2 + \omega^2}$
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Fourier Transform:

Frequency Shifting

If

$$x(t) \Longleftrightarrow X(\omega)$$

then

$$x(t)e^{j\omega_0 t} \Longleftrightarrow X(\omega - \omega_0)$$

Proof. By definition,

$$\mathcal{F}[x(t)e^{j\omega_0 t}] = \int_{-\infty}^{\infty} x(t)e^{j\omega_0 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t)e^{-j(\omega - \omega_0)t} dt = X(\omega - \omega_0)$$



Fourier Transform:

Modulation property:

Modulation property is an interesting consequence of the frequency shifting property combined with the linearity of the Fourier transform.

Modulation

Modulation property

For a transform pair

$$x(t) \xleftrightarrow{\mathcal{F}} X(\omega)$$

it can be shown that

$$x(t) \cos(\omega_0 t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

and

$$x(t) \sin(\omega_0 t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [X(\omega - \omega_0) e^{-j\pi/2} + X(\omega + \omega_0) e^{j\pi/2}]$$



Fourier Transform:

Modulation

$$x(t) \cos \omega_0 t = \frac{1}{2} [x(t) e^{j\omega_0 t} + x(t) e^{-j\omega_0 t}]$$

$$x(t) \cos \omega_0 t \iff \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

$$x(t) e^{j\omega_0 t} \iff X(\omega - \omega_0)$$

$$x(t) e^{-j\omega_0 t} \iff X(\omega + \omega_0)$$

This result shows that the multiplication of a signal $x(t)$ by a sinusoid of frequency ω_0 shifts the spectrum $X(\omega)$ by $\pm\omega_0$

Fourier Transform:

EXAMPLE 7.15 Spectral Shifting by Amplitude Modulation

Find and sketch the Fourier transform of the modulated signal $x(t) \cos 10t$ in which $x(t)$ is a gate pulse $\text{rect}(t/4)$, as illustrated in Fig. 7.26a.

From pair 17 of Table 7.1, we find $\text{rect}(t/4) \iff 4 \text{sinc}(2\omega)$, which is depicted in Fig. 7.26b. From Eq. (7.32) it follows that

$$x(t) \cos 10t \iff \frac{1}{2}[X(\omega + 10) + X(\omega - 10)]$$

In this case, $X(\omega) = 4 \text{sinc}(2\omega)$. Therefore,

$$x(t) \cos 10t \iff 2 \text{sinc}[2(\omega + 10)] + 2 \text{sinc}[2(\omega - 10)]$$

The spectrum (Fig. 7.26c) of $x(t) \cos 10t$ is obtained by shifting $X(\omega)$ in Fig. 7.26b to the left by 10 and also to the right by 10, and then multiplying it by 0.5, as depicted in Fig. 7.26d.

$$\text{rect}\left(\frac{t}{\tau}\right) \iff \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$\omega\tau/2 = \pm n\pi$$

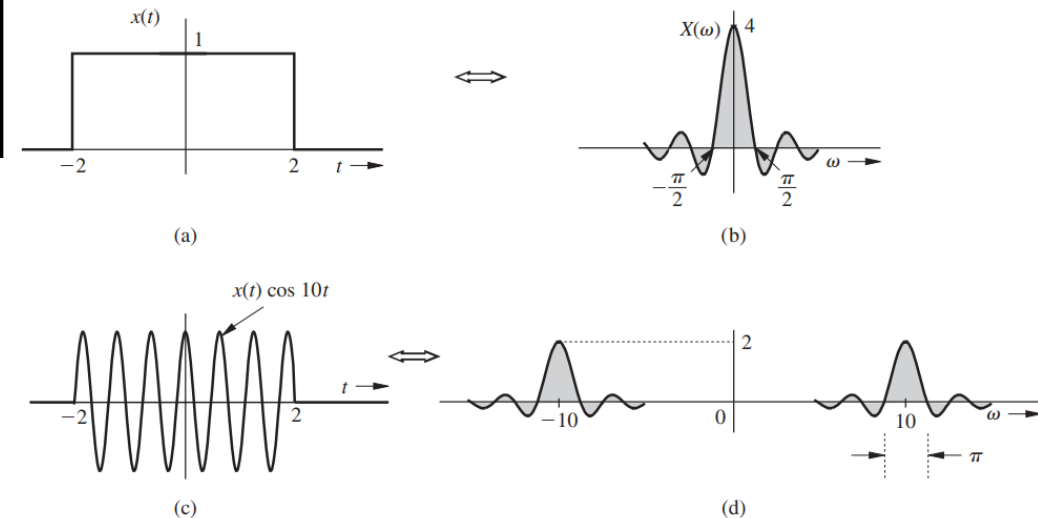


Figure 7.26 An example of spectral shifting by amplitude modulation.



Quiz-5:

Topic: Fourier Transform

Date: Monday, 20th April – 30min of class timings

For Practice:

1. Solve all problems in the lecture slide- Find FT of basic signals (application of FT formula, integration).
2. Use FT tables (basic FT pair and properties), find FT/IFT expression and its plot -----HW4 - Q3, 5
3. Match the time domain signal with the given FT plotHW4- Q 4
4. Prove all FT properties (read the book) and solve the practice problem set 05.

Thanks!